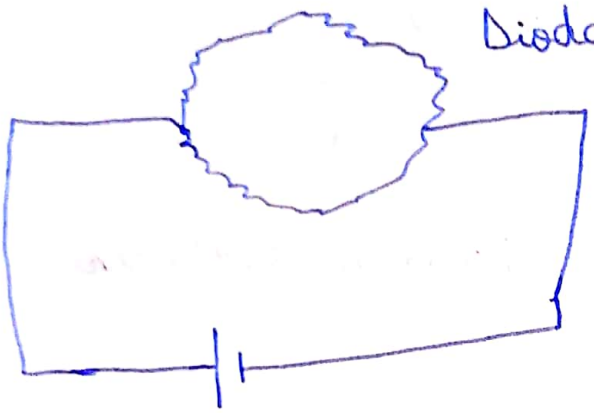
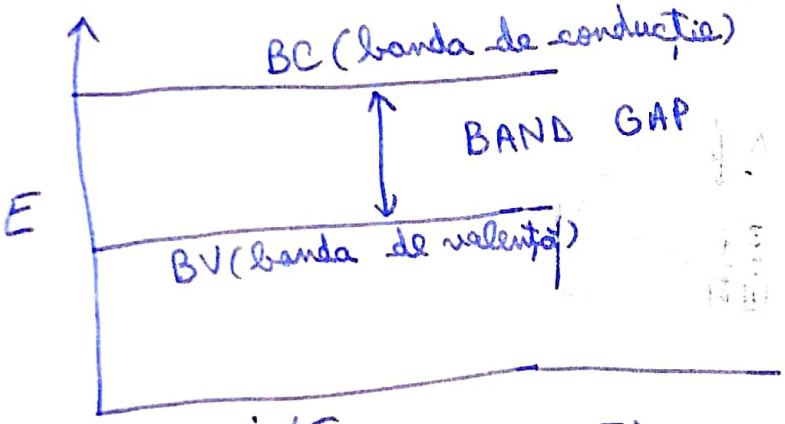
13.X.2022

DEEA
Curs 3

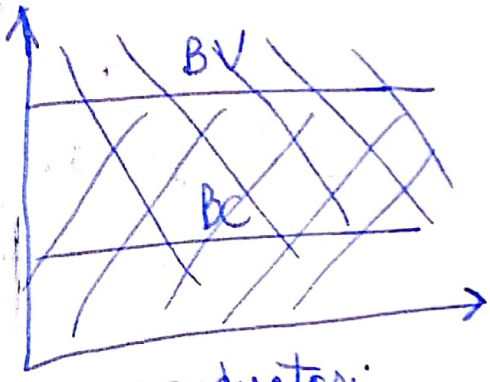
Semiconductori
Dioda Semiconductoare



- Izolatori: cauciuc, plastic, sticlă, aer — $1.5 eV < E_{BG}$ (mare)
- Conductori: metale
- Semiconductori: siliciu, germaniu — $E_{BG} \simeq 1 eV$



i / S

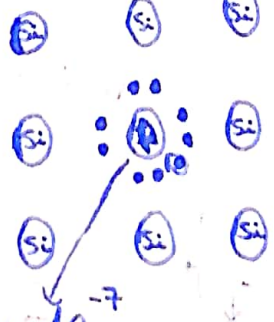


conductori

$E = \frac{U}{D}$

$F \sim E \cdot q_0$

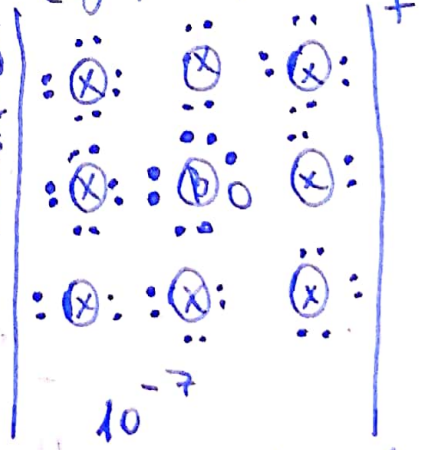
(grupa V)
semicond. N



10^{-7}

curent de electroni
(purtători negativi)

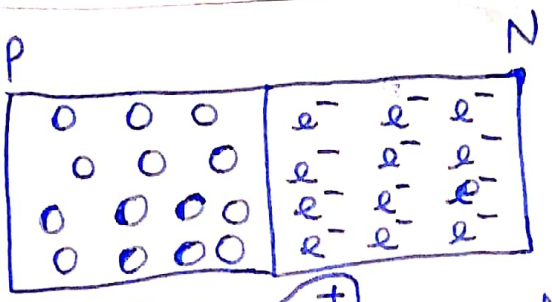
(grupa III)
semicond. P

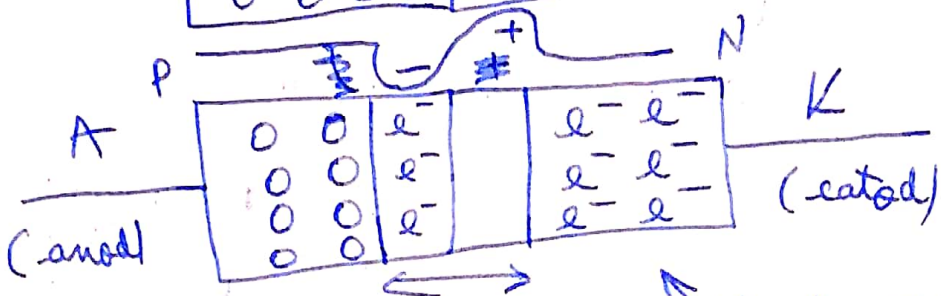


10^{-7}

curent de goluri
(purtători pozitivi)

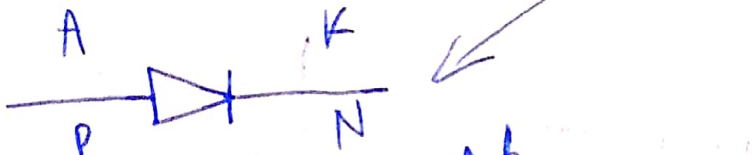
Scanned with CamScanner



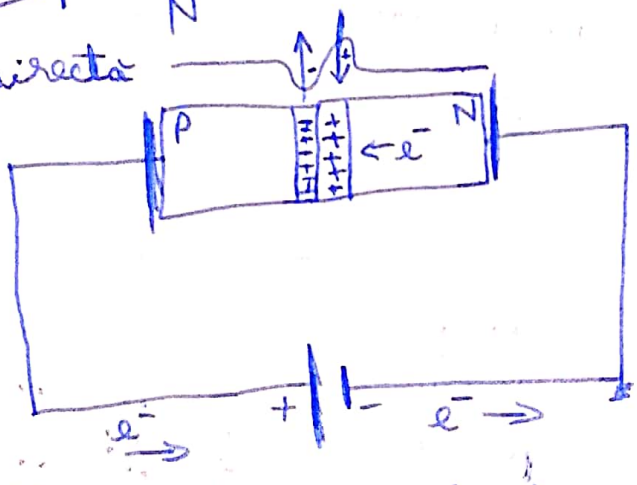


zonă fără purtători liberi

Diodă semiconductoare

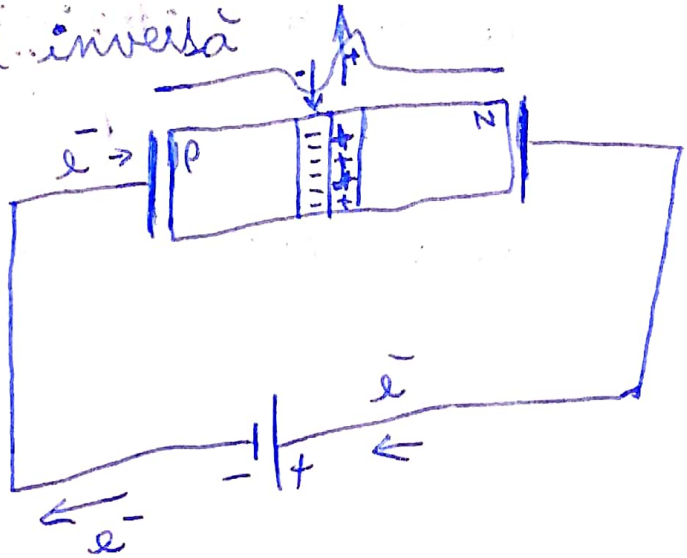


Polarizare directă



curent de recombinare

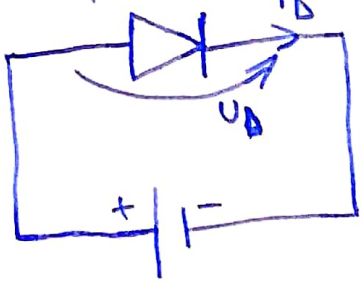
Polarizare inversă



curent de generare

Scanned with CamScanner

D.R. A K i_D

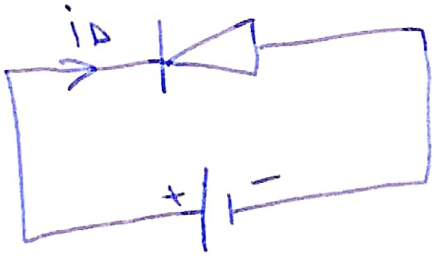


$$i_D = i_S\left(e^{\frac{U_D}{V_T}} - 1\right), \quad U_D > 0$$

$$V_T = \frac{KT}{q_0} \approx 26\,mV \approx 25\,mV \text{ (pt. calcule)}$$

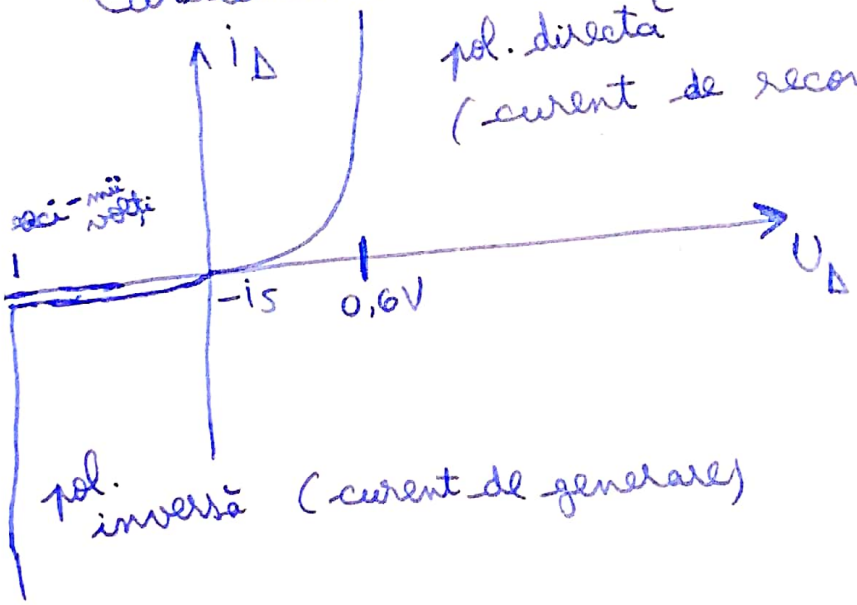
$$i_S \approx 10^{-17}\,A$$

POL. INV.



$$i_D = i_S, \quad U_D < 0$$

Caracteristica statică



Scanned with CamScanner

18.X.2022

DEEA
Curs 4

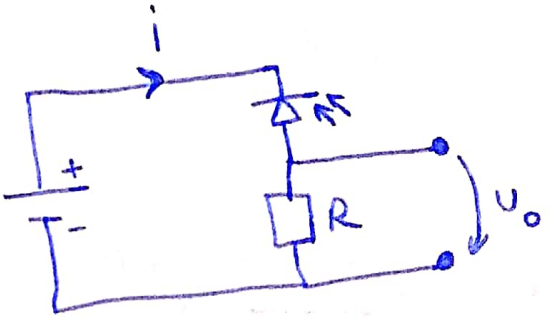
Tipuri constructive:

1)  D. obișnuită

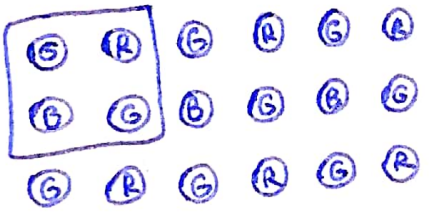
2)  PHOTODIODĂ (pol. inv.; lumină → generare)

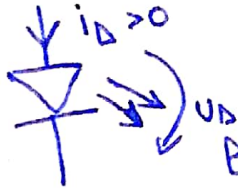
$i_D \sim E_v \cdot A$

Lux
W/m^2
↳ Lumină/m^2



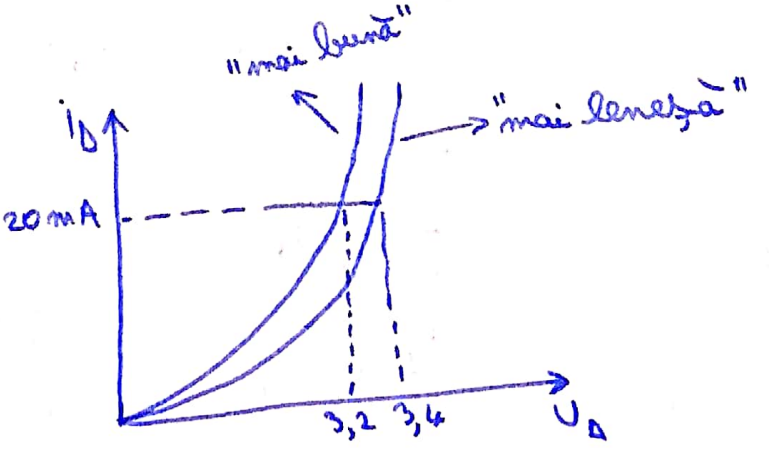
$U_o = i_D R = i_S R \sim E_v A R$



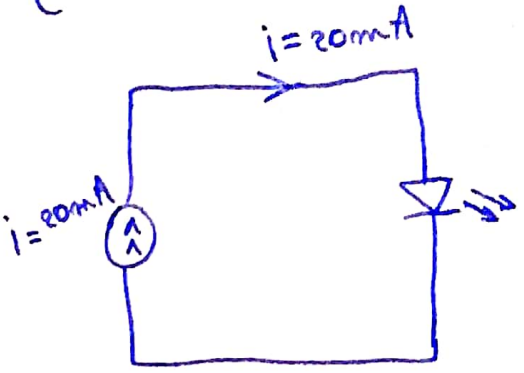
3)  L.E.D

Pol. directă

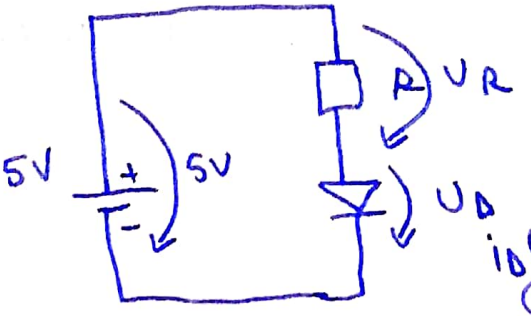
RECOMBINARE
E-G → Lumină



V_F	3,2	3,4
$i_{D\,MAX}$	20mA	



Cale. pt. $V_{F\,min}$

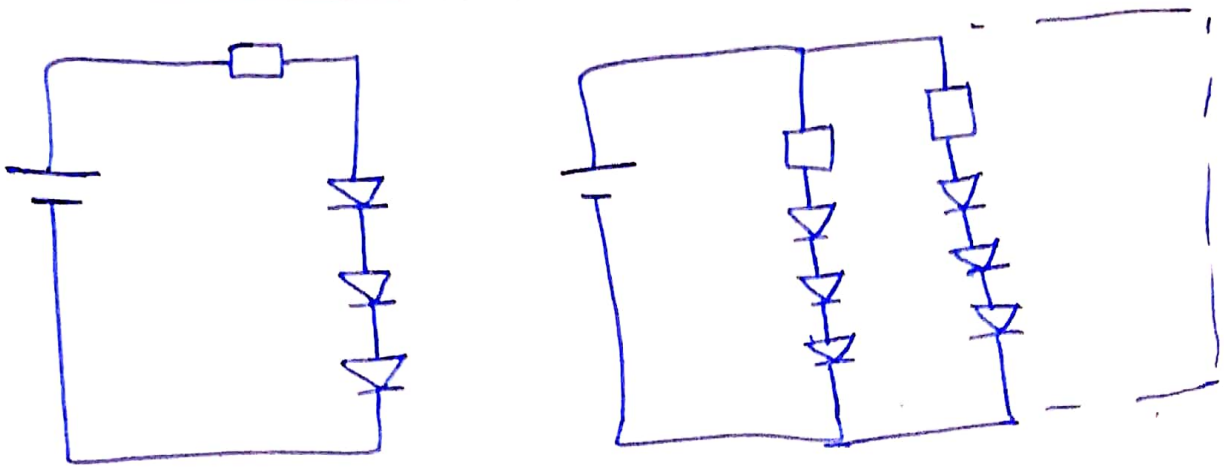


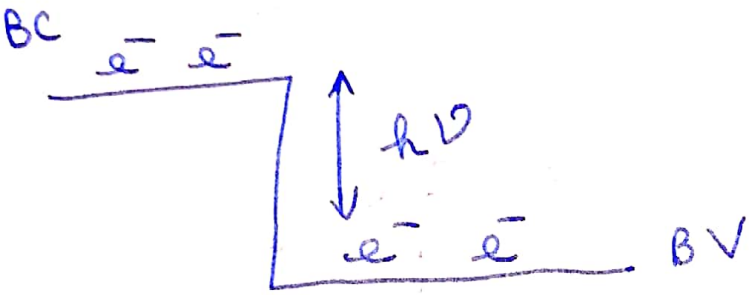
$E = U_D + i_D R$

$\Rightarrow R_1 = \frac{5V - 3,2V}{20mA} = 0,09K$

$i_D^{(V_F=3,4V)} = \frac{5V - 3,4V}{0,09K} = 17,7mA$

Scanned with CamScanner



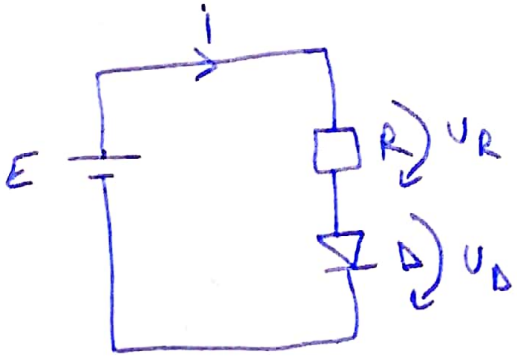


LED ALB



η : 3% INC
15% LED

MODELE MATEMATICE
PT. DIODE
(APROXIMAȚII)



$E = U_R + U_D$

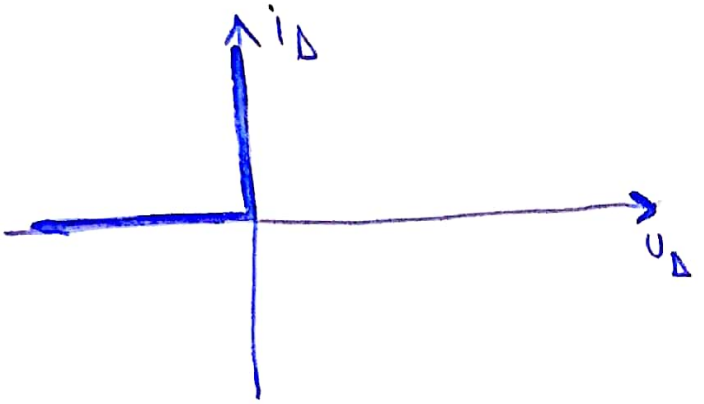
$E = iR + U_D$

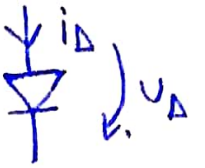
$E = R i_S \left(e^{\frac{U_D}{U_T}} - 1\right) + U_D$

TRANSCEDENTALĂ

Scanned with CamScanner

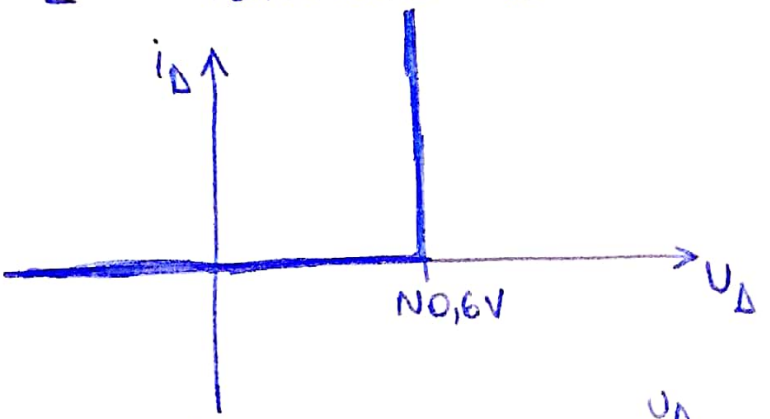
APROX. 1 – DIODA IDEALĂ





$$\begin{cases} i_D = 0 & U_D < 0 \\ U_D = 0 & i_D > 0 \end{cases}$$

APROX. 2 – TENSIUNE CONSTANTĂ



$$\begin{cases} i_D = 0 & U_D < 0{,}6 \\ U_D = 0{,}6 & i_D > 0 \end{cases}$$

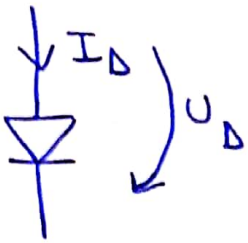
$$i_D = i_S \left(e^{\frac{U_D}{V_T}} - 1 \right) \approx i_S e^{\frac{U_D}{V_T}} \qquad V_T \approx 25\,mV$$

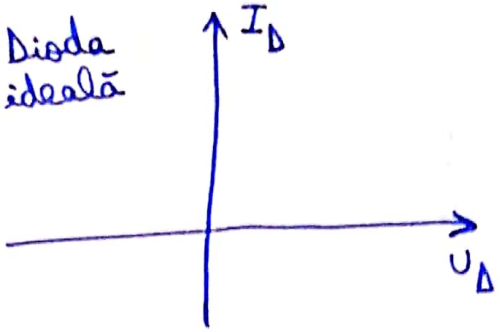
U_D	i_D
400mV	1μA
425mV	
500mV	
600mV	
700mV	

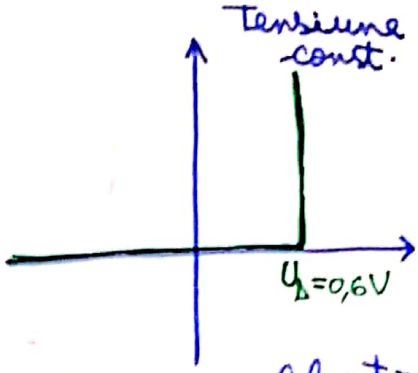
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Curs 5 - DEEA

20.X.2022



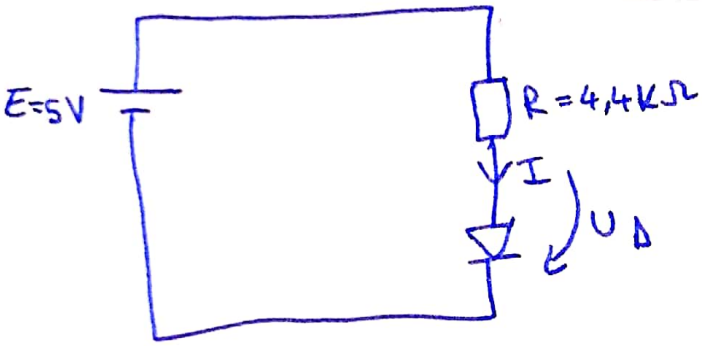




$$\begin{cases} U_D < 0, \; I_D = 0 \\ I_D > 0, \; U_D = 0 \end{cases}$$

$U_D < 0,6V$ $I_D = 0$ blocată

$I_D > 0$ $U_D = 0,6V$ conducție



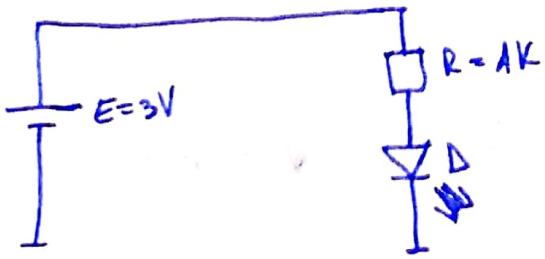
$U_D = 0,6V$ (conducție)

$I = ?$

Pp. că dioda este în cond.

KVL KII $\quad E = IR + U_D$

$$I = \frac{E - U_D}{R} = 1mA > 0 \quad \text{(PP adev)}$$



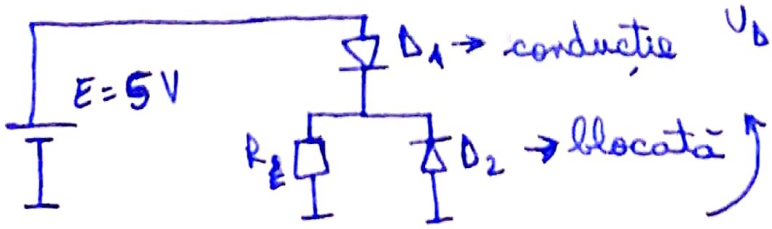
PP că Dioda e în cond. $\Rightarrow U_D = 3,3V$

$E = IR + U_D$

$$I = \frac{E - U_D}{R} = -0,3mA$$

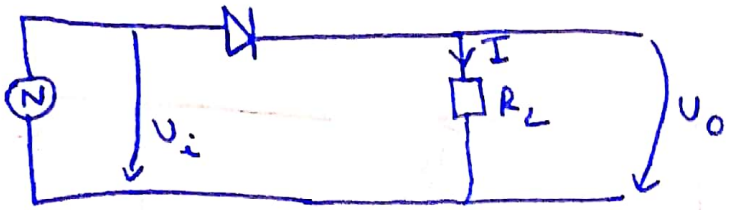
(PP greșit)

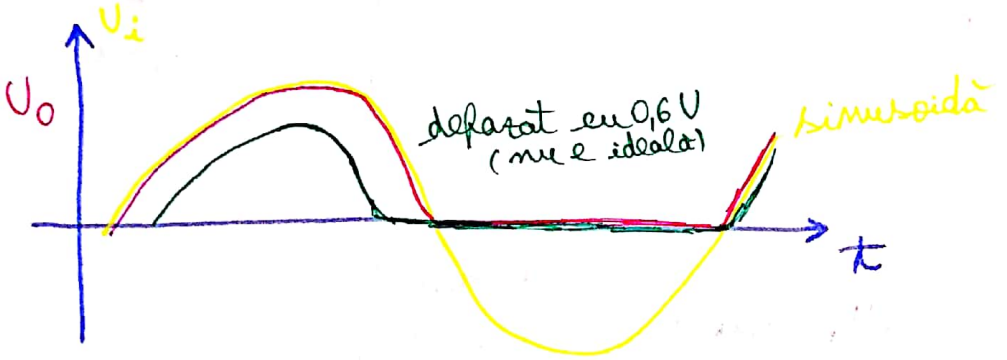
$\Rightarrow$ Dioda e blocată $\Rightarrow i = 0A$

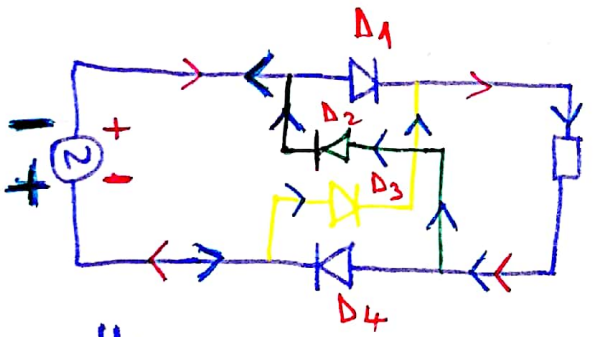


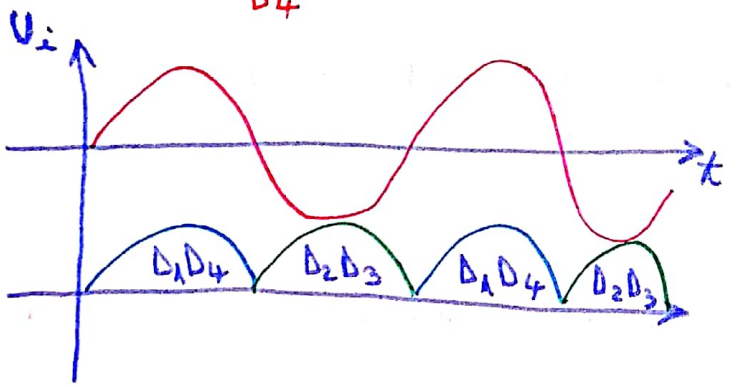
Scanned with CamScanner

REDRESOR MONOALTERNANT

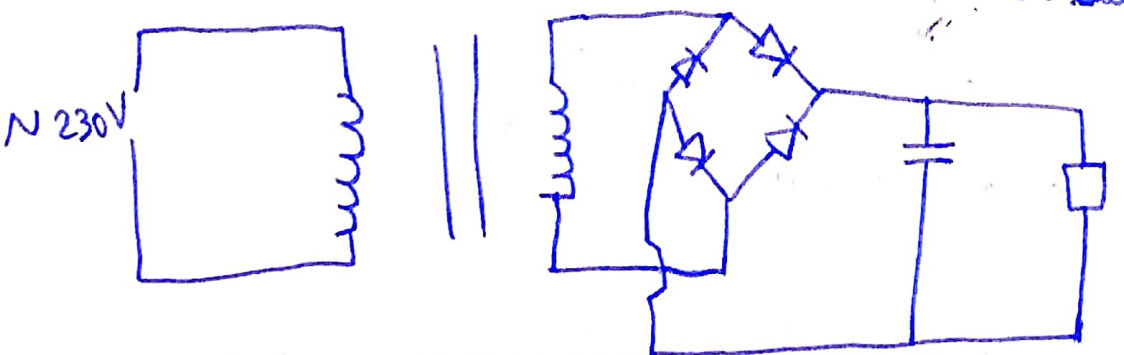


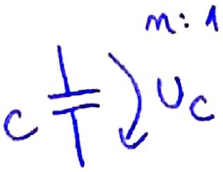






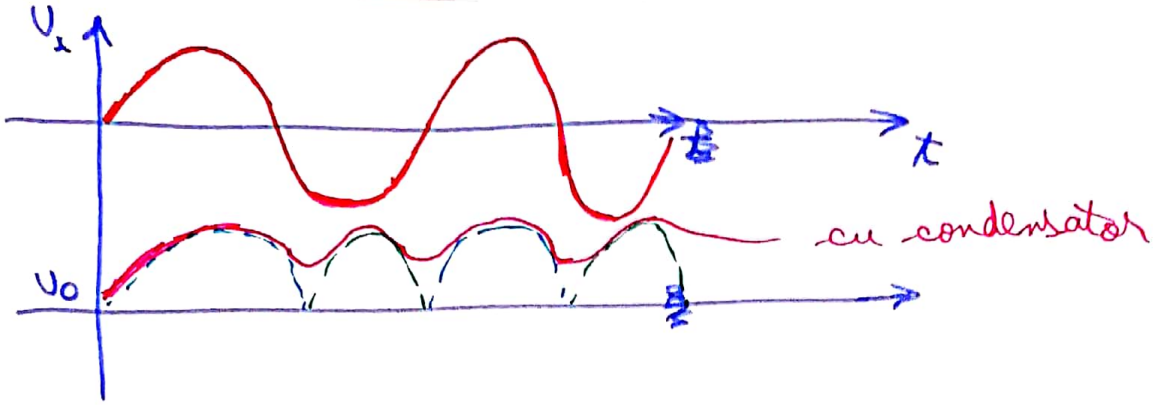
Redresor alternant



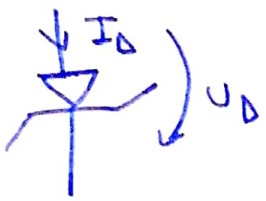


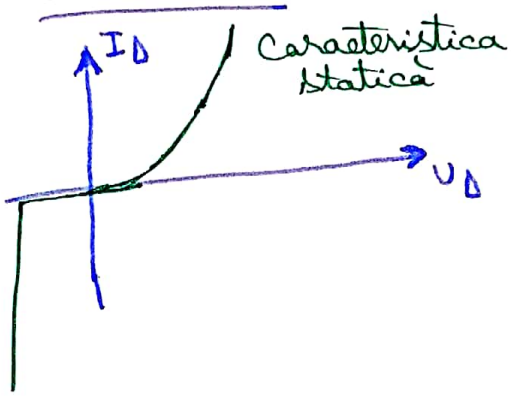
$$I_C = C\frac{dU_C}{dt}$$

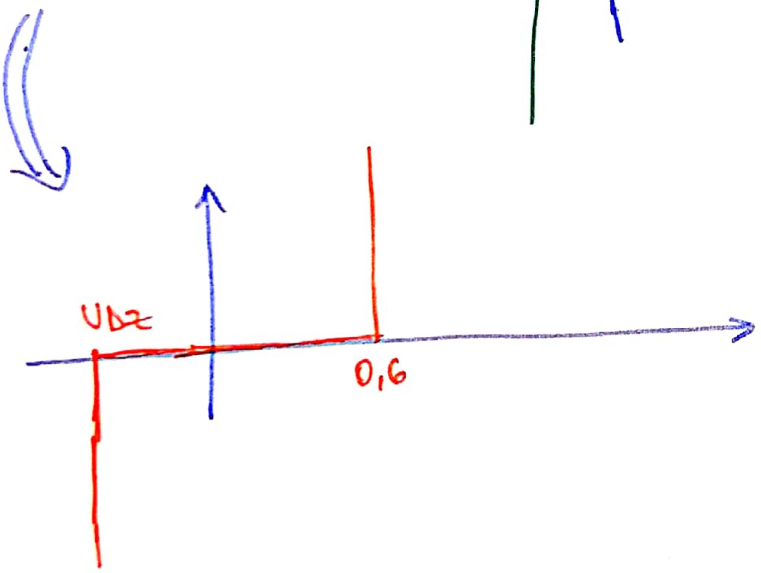
Scanned with CamScanner



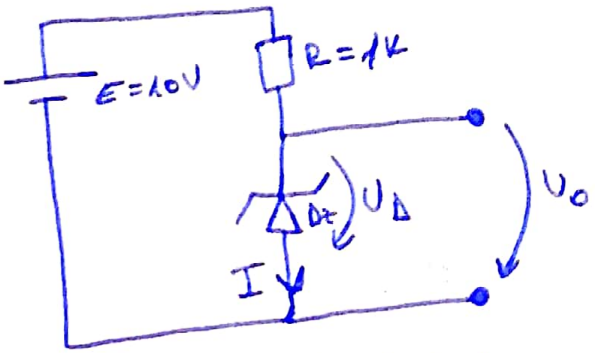
Dioda Zener







$$\begin{cases} I_D > 0,\ U_D = 0{,}6\ V & \text{conducție} \\ -U_{Dz} < U_D < 0{,}6,\ I_D = 0 & \text{blocat} \\ I_D < 0,\ U_D = -U_{Dz} & \text{conducție inversă} \end{cases}$$



Aici curentul trece invers prin diodă

$U_D = 0{,}6V$ (cond. diodă)

$U_{Dz} = 3V$

$U_0 = ?$

PP. că Dz = cond. inv.

$I_D > 0,\ U_D = U_{Dz}$??

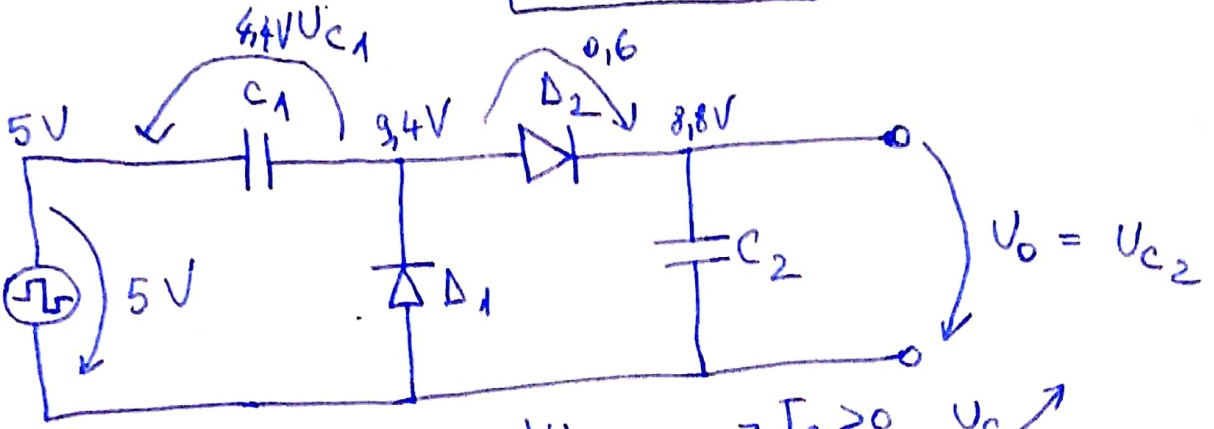
$E = RI + U_{Dz} \Rightarrow I = \frac{E - U_{Dz}}{R} = 5\,mA$

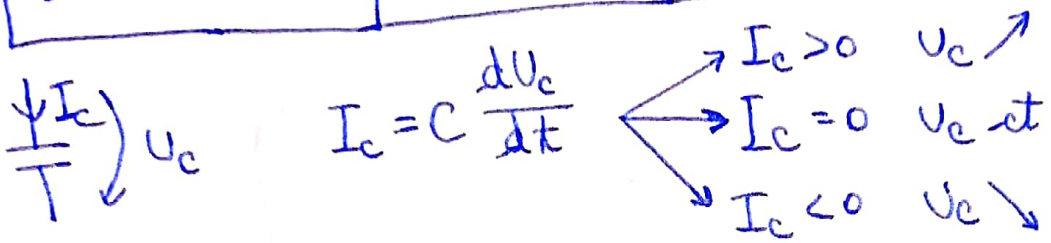
$U_0 = U_{Dz} = 5V$

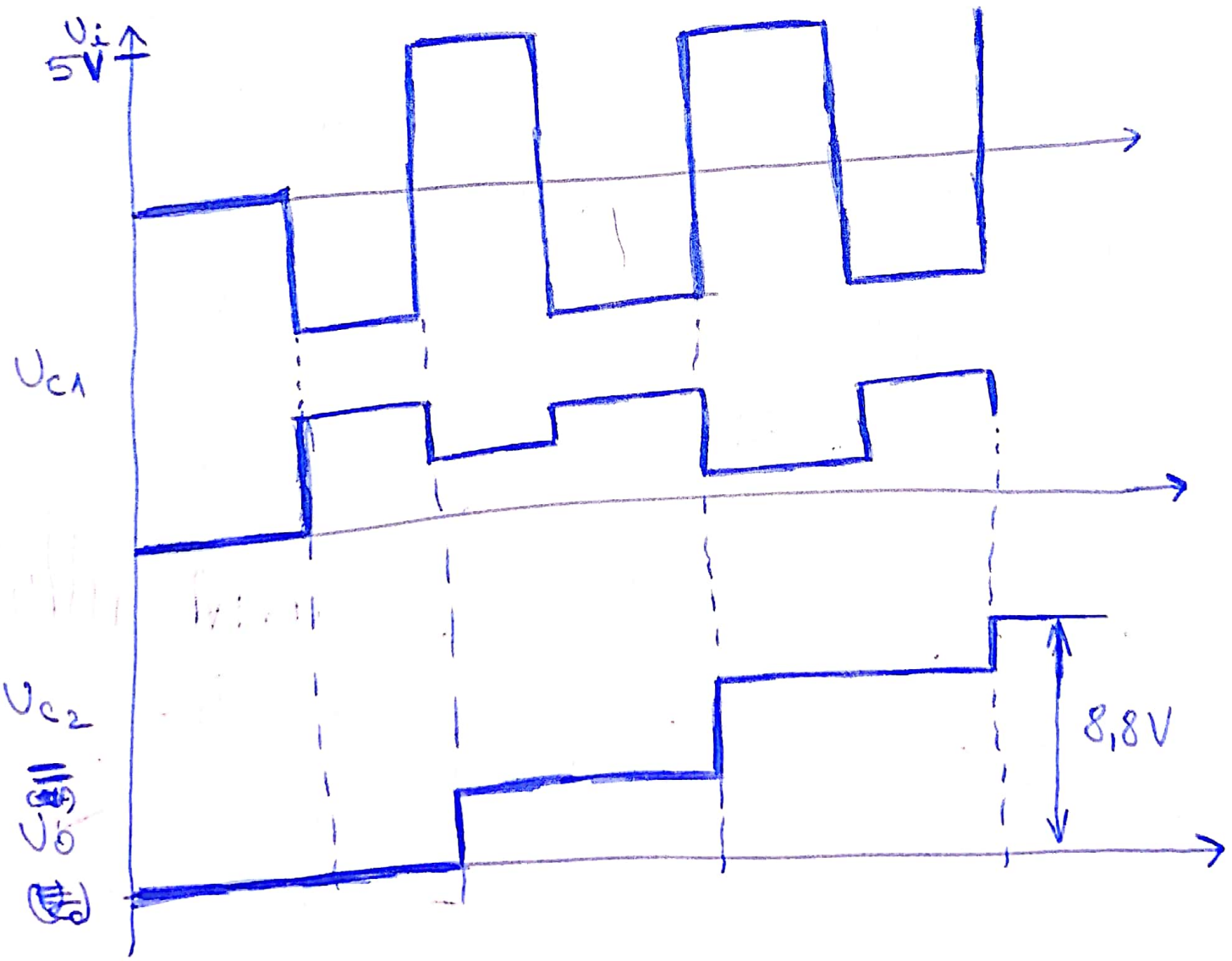
Scanned with CamScanner

Curs 6 DEEA

27.X.2022



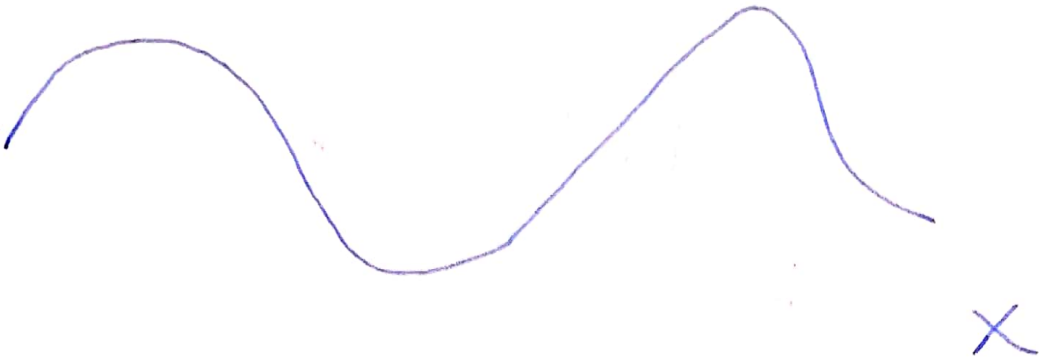




Scanned with CamScanner

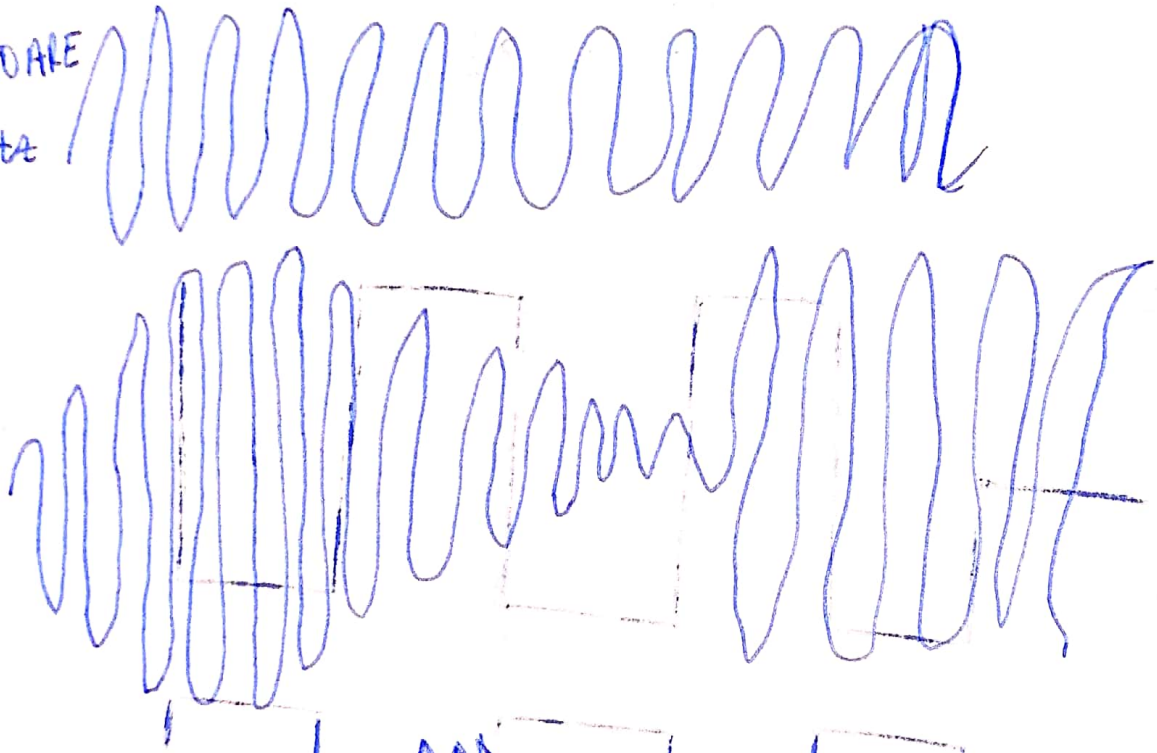
AUDIO

20Hz - 20KHz

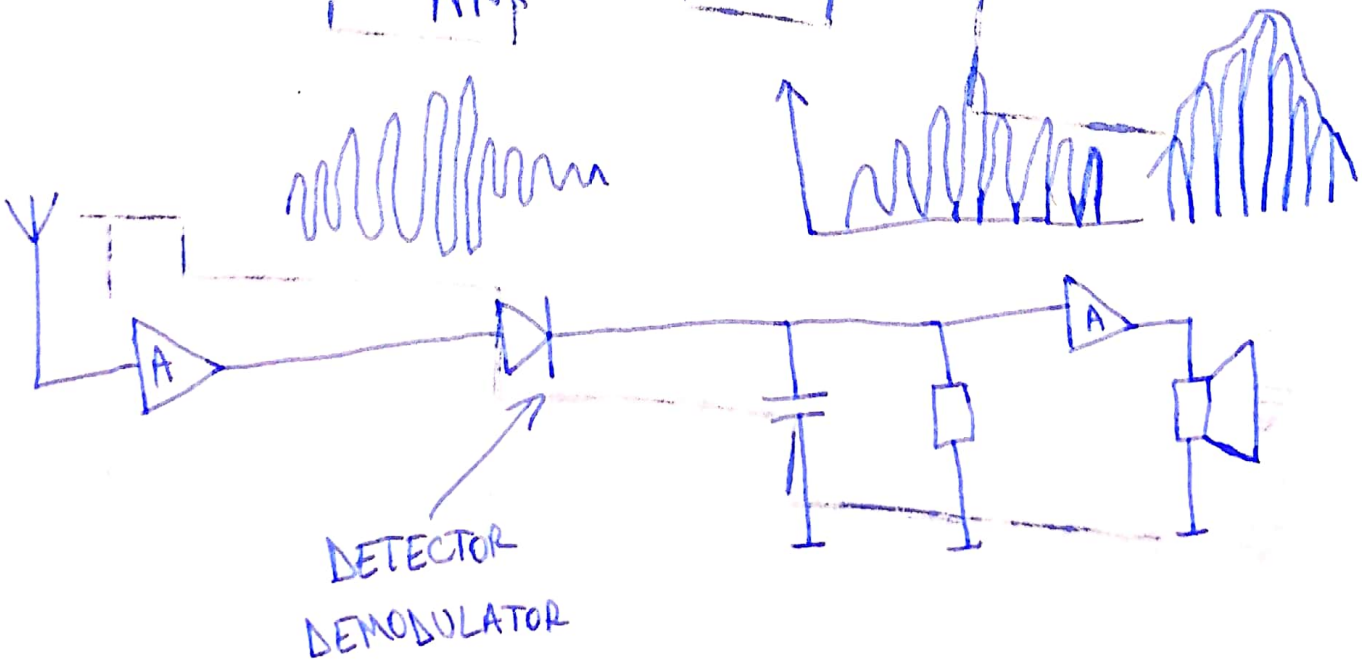


PURTĂTOARE

800KHz



AM

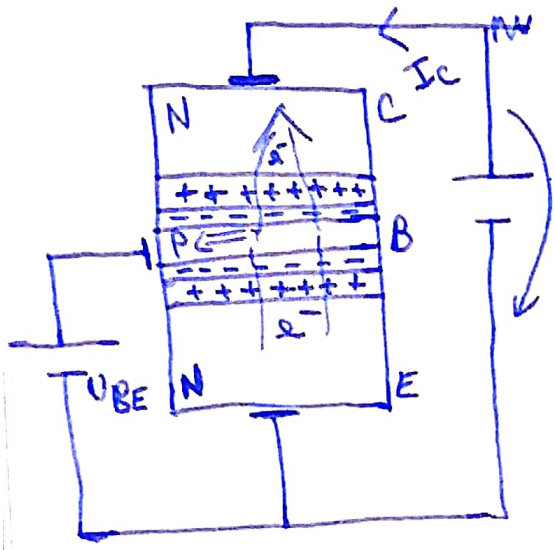


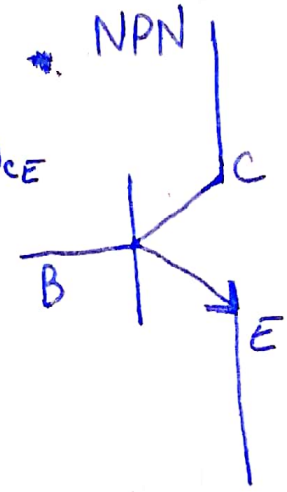
900 MHz → 3,5 GHz

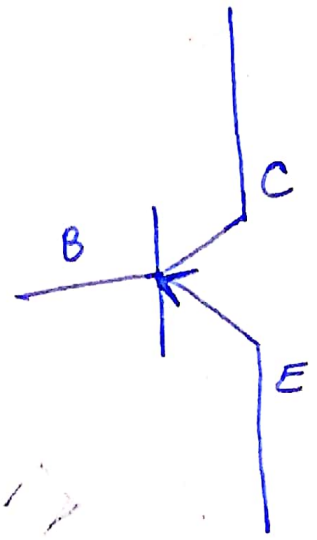
Scanned with CamScanner

1951

Tranzistor bipolar







$\begin{cases} U_{BE} > 0 \\ U_{CE} > U_{BE} \end{cases}$

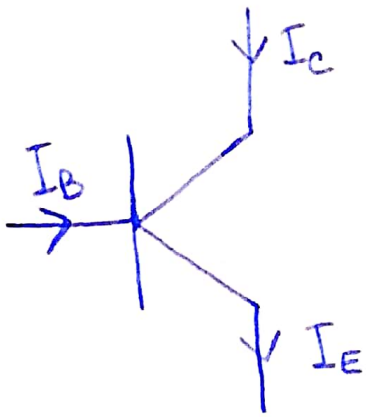
REGIUNE ACTIVĂ NORMALĂ

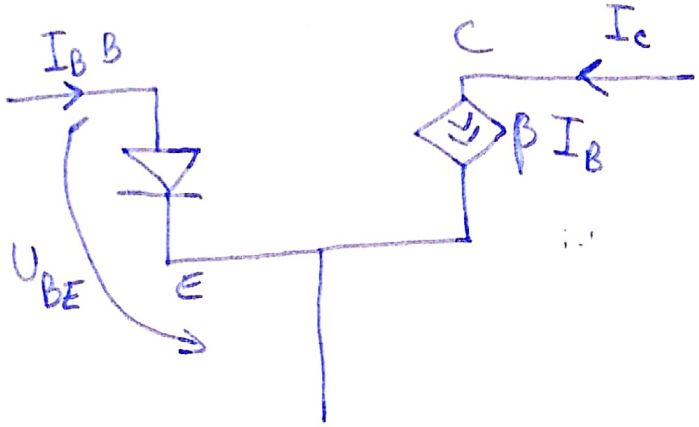
$I_C \gg I_B$

$I_E = I_B + I_C$

$I_B \overset{N}{\sim} I_C$

$I_C = \beta I_B$ β N SUTE





$I_C = \beta I_B$ $I_B \ll I_C$

$I_E = I_C + I_B \overset{N}{=} I_C$

$U_{BE} \overset{N}{=} 0{,}6V$

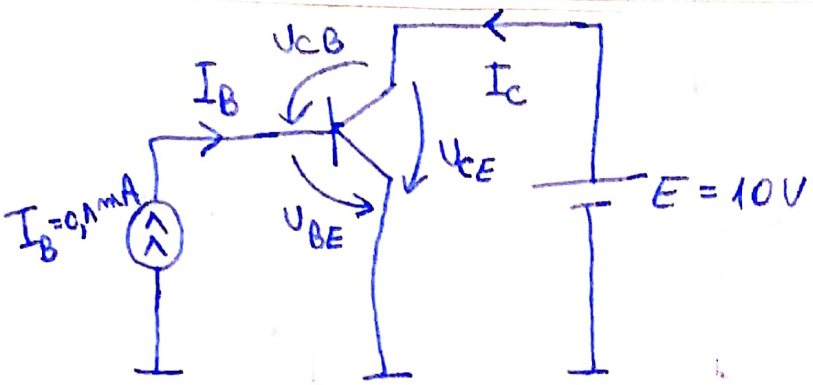
BE POL DIRECTĂ

$U_{BE} = 0{,}6$

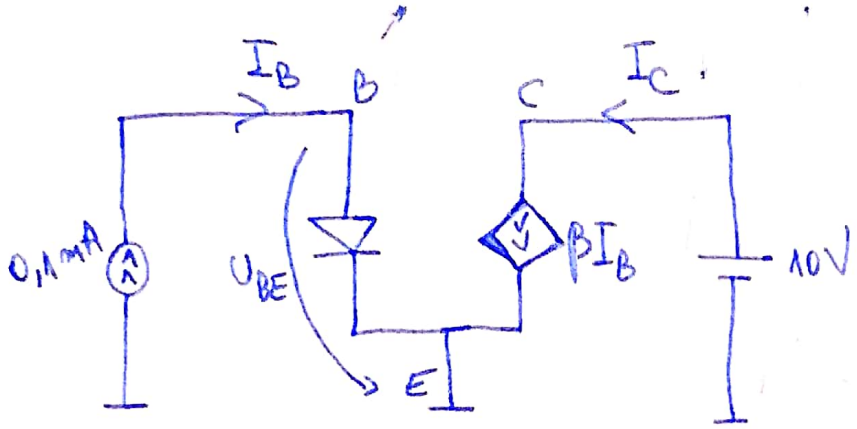
CB POL INV.

$U_{CB} > 0$

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$\beta = 100$



$I_B = 0,1\,mA \Rightarrow U_{BE} = 0,6\,V$

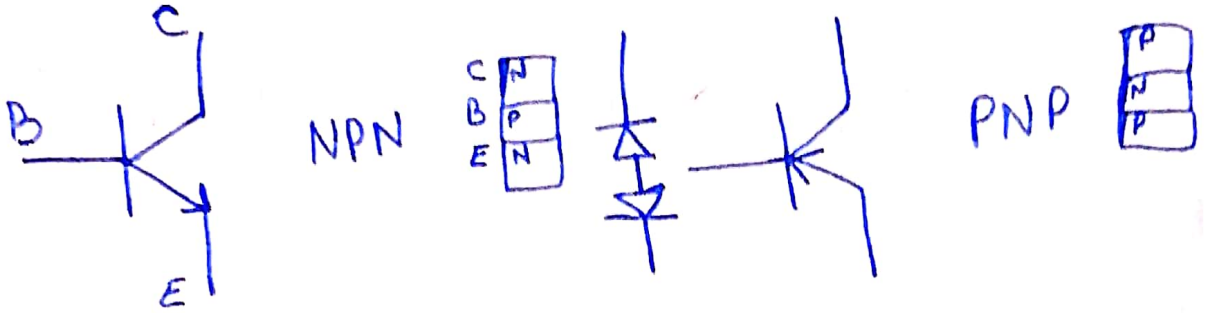
$I_C = \beta I_B = 100 \cdot 0,1\,mA = 10\,mA$

$U_{CE} = E = 10\,V \qquad U_{CB} = U_{CE} - U_{BE} = 9,4\,V$

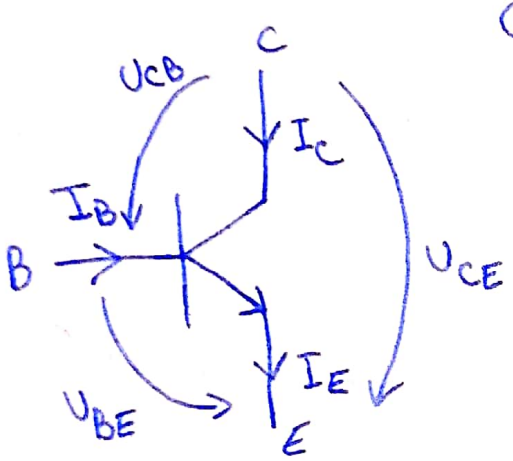
Scanned with CamScanner

1.XI.2022

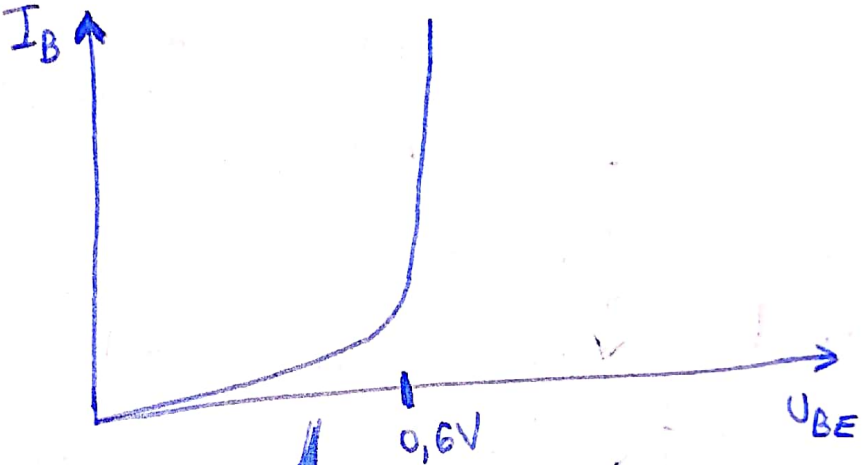
Curs 7 - DEEA

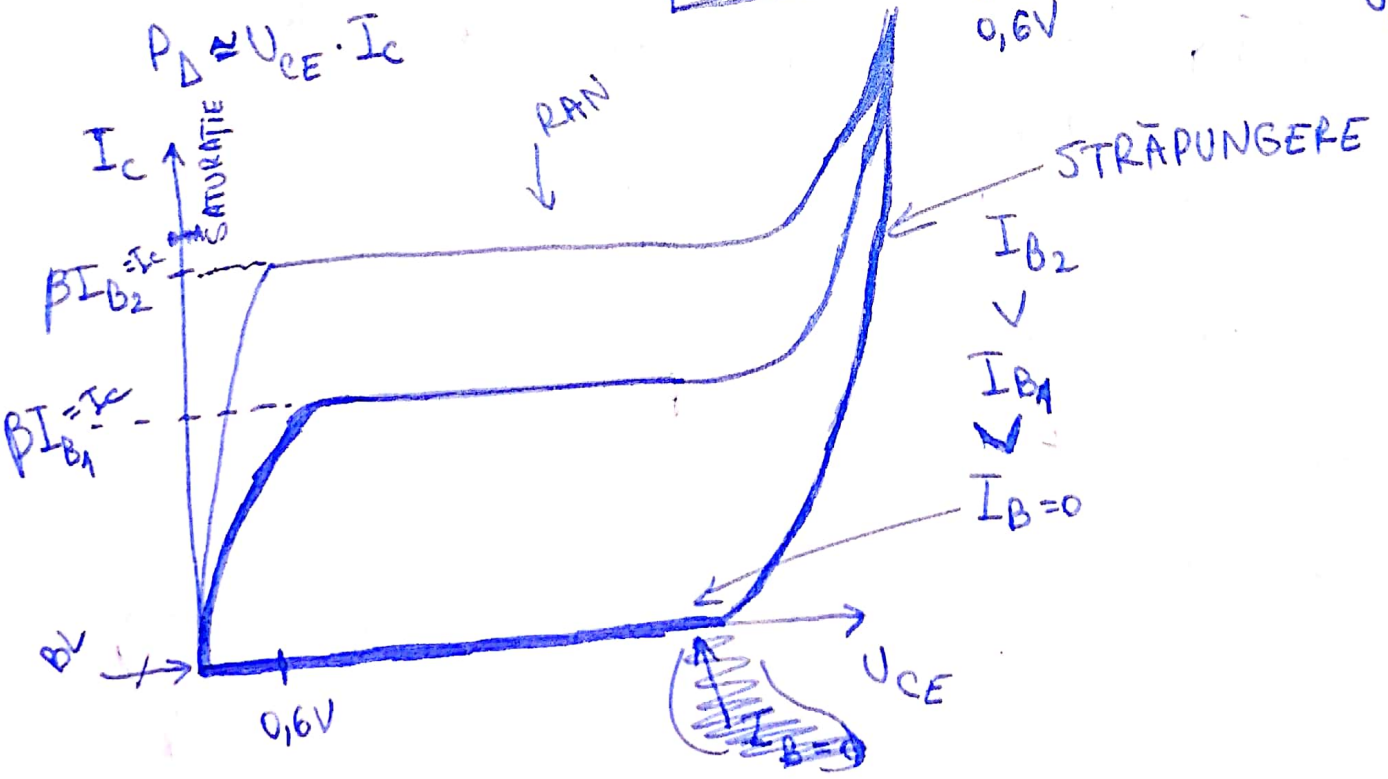


CARACTERISTICI STATICE



$P_D \approx U_{CE} \cdot I_C$





Scanned with CamScanner

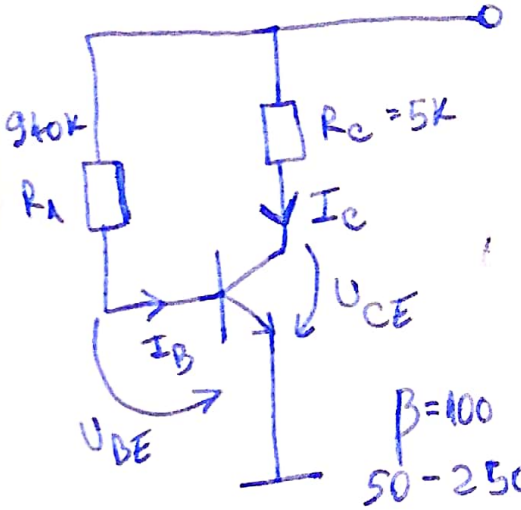
	RAN (regiunea activă mat.) DIRECT INVERS	SAT (saturat)	RAI (—||— —||— inv.)	BLOCAT
NPN	$U_{BE}>0$ $U_{CB}>0$	$U_{BE}>0$ $U_{CB}<0$	$U_{BE}<0$ $U_{CB}<0$	$U_{BE}<0$ $U_{CB}>0$
PNP	$U_{EB}>0$ $U_{BC}>0$	$U_{EB}>0$ $U_{BC}<0$	$U_{EB}<0$ $U_{BC}<0$	$U_{EB}<0$ $U_{BC}>0$
				$I_C=0$

$I_C=\beta I_B$

$U_{BE}\cong 0,6V$

β = zeci-sute

POLARIZAREA TRANZISTORULUI



Pt. T. RAN $\Rightarrow I_C=\beta I_B$ $U_{BE}\cong 0,6V$

Punct static de funcționare

I_C U_{CE}

$V_{cc}=R_1 I_B+U_{BE}$

$I_B=\frac{V_{cc}-U_{BE}}{R_1}\cong\frac{9,4}{940K}=0,01mA$

$I_C=\beta I_B=1mA$

$V_{cc}=U_{CE}+I_cR_c \Rightarrow U_{CE}=V_{cc}-I_cR_c=10V-5V=5V$

$U_{BE}=0,6$

$\beta=200$ $I_B=0,01mA$ $I_C=2mA$ $U_{CE}=0$ NU E ÎN RAN

$\beta=300 \Rightarrow U_{CE}\cong 0$

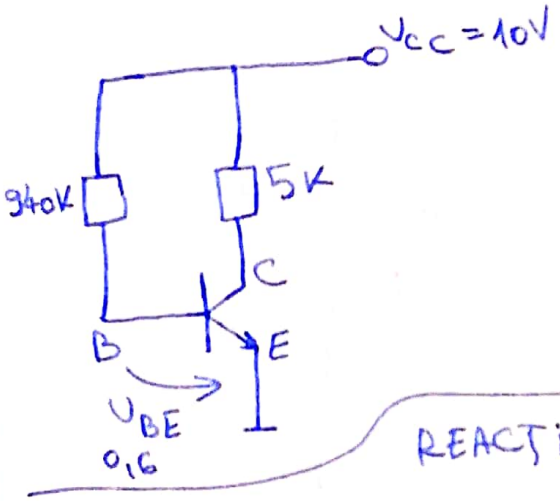
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3.XI.2022

Curs 8 - DEEA

E.A. $\Rightarrow$ M.F. $\not\sim$ M.E. $\Rightarrow$ T - R.A.N. — POLARIZĂM $\downarrow$ P.S.F. (I_C, V_{CE})

$$\begin{cases} I_C = \beta I_B \\ V_{BE} \approx 0{,}6V \end{cases}$$

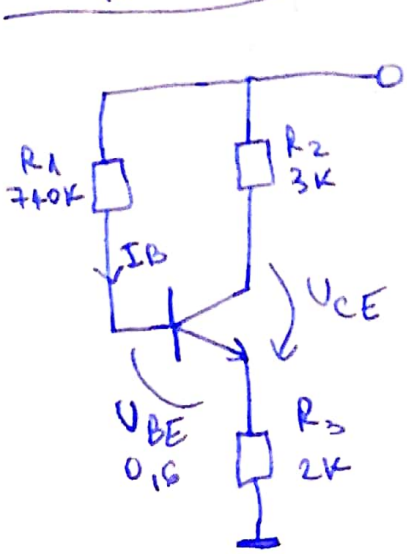


$I_B = 0{,}01\,mA$

$\beta = 100 \Rightarrow I_C = 1\,mA$ RAN
$V_{CE} = 5V$

$\beta = 200 \Rightarrow I = 2\,mA$ ~~RAN~~
$V_{CE} = 0V$

REACȚIE NEGATIVĂ



$$V_{CC} = R_1 I_B + V_{BE} + I_E R_3$$

~~REACȚIE NEGATIVĂ~~

$$I_B = \frac{I_C}{\beta}$$

$$I_E = I_B + I_C = \frac{I_C}{\beta} + I_C \approx I_C$$

$$I_C = \frac{V_{CC} - V_{BE}}{R_3 + \frac{R_1}{\beta}} \qquad \beta = 100$$

$$I_C = \frac{9{,}4V}{2 + 7{,}4} = 1\,mA$$

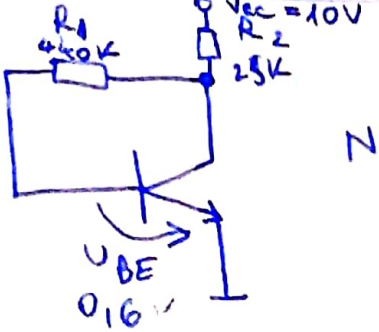
~~I_{CE}~~ $V_{CC} = I_C R_2 + V_{CE} + I_E R_3 \approx V_{CE} + I_C (R_2 + R_3)$

$V_{CE} = 10V - 5V = 5V$

$\beta = 200 \quad I_C = \frac{9{,}4V}{2 + 3{,}7} = 1{,}64\,mA$

$V_{CE} = 10V - 5 \cdot 1{,}64\,mA = 4{,}8V$

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NPN

$$V_{cc} = R_2(I_c + I_B) + R_1 I_B + V_{BE}$$

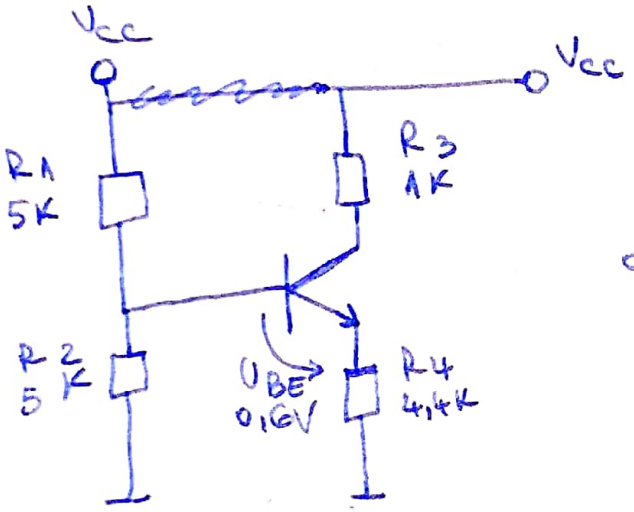
$\beta = 100$

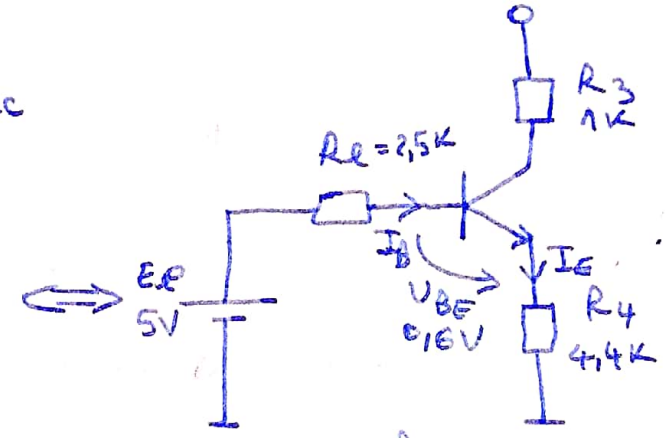
$$I_c = \frac{V_{cc} - V_{BE}}{R_2 + \frac{R_2}{\beta} + \frac{R_1}{\beta}} = \frac{9,4\ V}{25K + \frac{25K}{100} + \frac{440K}{100}} =$$

$$= 1\ mA$$

$\beta = 200$ $\quad I_c = \frac{9,4\ V}{7,2K} = 1,3\ mA$ $\qquad$ PSF (I_c, V_{CE})

Polarizare 4R





$$E_e = V_{cc} \frac{R_2}{R_1 + R_2} = 5\ V$$

$$R_e = R_1 \| R_2 = 2,5\ K$$

P.S.F.

$$E_e = I_B R_e + V_{BE} + I_E R_4$$

$$E_e = R_e \frac{I_c}{\beta} + V_{BE} + \left(\frac{I_c}{\beta} + I_c\right) R_4$$

$$I_c = \frac{5\ V - 0,6\ V}{\frac{2,5K}{100} + \frac{4,4K}{100} + 4,4K} \cong 1\ mA \quad (\beta = 100)$$

$\beta = 200$ $\quad I_c \cong 1\ mA$

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Curs 9 - DEEA

10.XI.2022

RAN

BE pol. directă

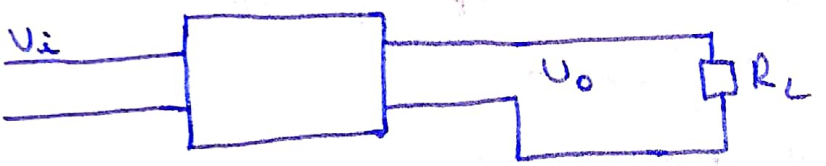
CB pol. indirectă

$I_C = \beta I_B$

Componente → polarizare → P.J.F. (~~pct~~)

$I_C, U_{CE} > U_{BE}$ (NPN)

Stabilizator de tensiune serie



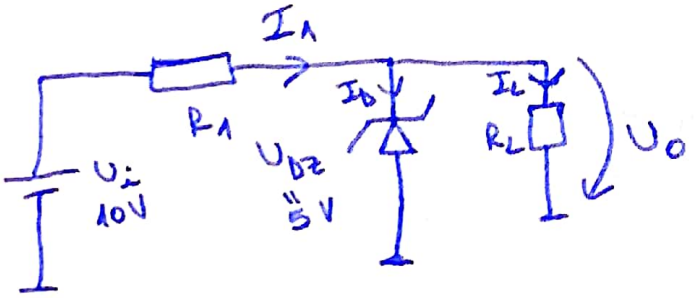
$U_o = ct$, $U_o \neq f(U_i)$, $U_o \neq f(R_L)$

$U_i = 100 - 270V$ ~ AC

$U_o = 19V / 3A$ DC

0-3A

STABILIZATOR PARALEL



$I_1 = \frac{U_i - U_D}{R_1} = 50mA$

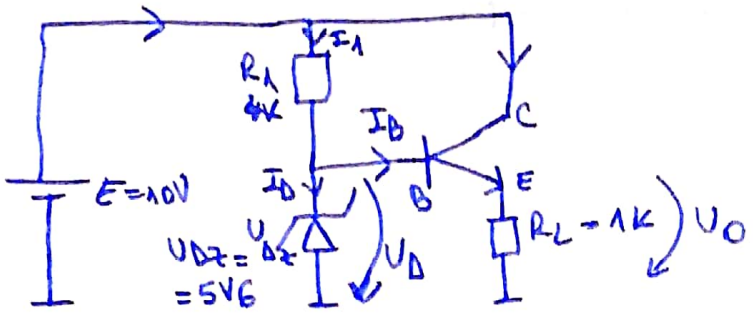
$I_1 = I_D + I_L$

$I_D > 0 \Rightarrow U_o = U_{Dz} = ct$

R_L	I_L	I_D	I_1
1K	5mA	45mA	50mA
0,5K	10mA	40mA	50mA
∞	0	50mA	50mA
~~10 Ω~~	~~50mA~~		

↑ nu mai stabilizează

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PP Tranz. în RAN, Dz cond. inversă ($I_D > 0$) $\Rightarrow U_D = U_{Dz} = 5,6\ V$

$I_C = \beta I_B$, $U_{BE} \approx 0,6\ V$

$U_D = U_{BE} + U_O \Rightarrow U_O = U_D - U_{BE} = 5V$

$I_1 = I_D + I_B$

$U_O = R_L I_E \approx R_L I_C \Rightarrow I_C = 5mA \Rightarrow I_B = \frac{I_C}{\beta} = 0,05\ mA$

$E = R_A I_1 + U_D \Rightarrow I_1 = \frac{4,4V}{4K4} = 1mA$

$I_D = 1mA - 0,05\ mA = 0,95\ mA > 0$

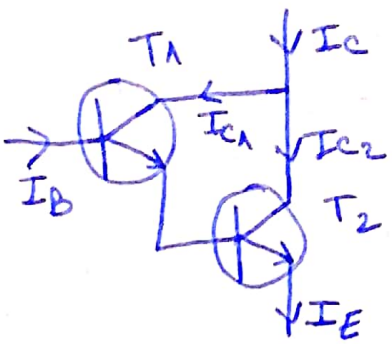
R_L	I_C	I_D	I_1	$I_i = I_1 + I_C \approx I_1 + I_L$
1K	5mA	0,95mA	1mA	6 mA
0,5K	10mA	0,9mA	1mA	11 mA
∞	0	1mA	1mA	1 mA

$I_{L\,max} = 100\ mA$

Bt. $I_{L\,max}$ m. mare $\Rightarrow R_1$ mai mic

TRANZISTOARE COMPUSE

- T. DARUNGTON



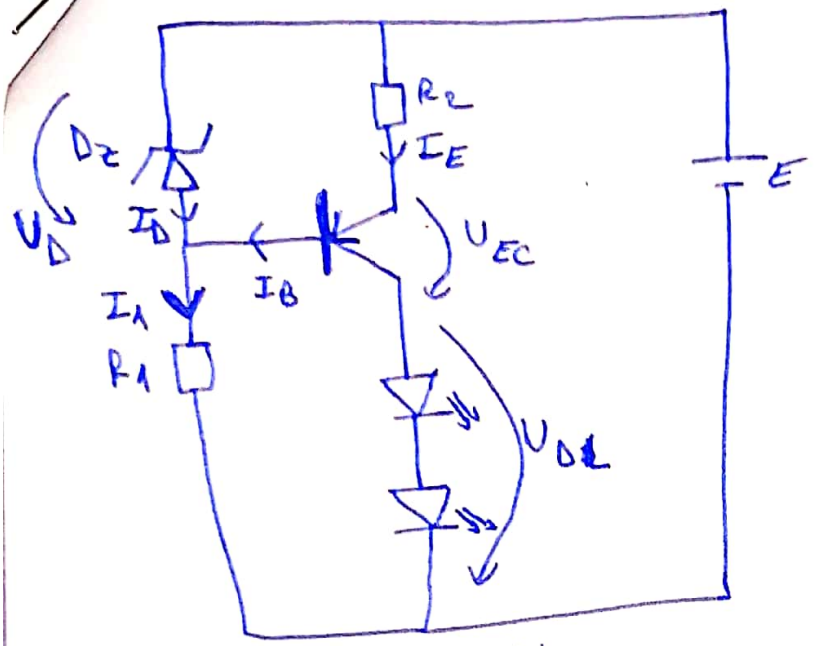
$I_C = I_{C_1} + I_{C_2}$

$I_{B_2} = I_{E_1} = I_{B_1}(\beta + 1) = I_B(\beta + 1)$

$I_C = \beta I_{B_1} + \beta I_{B_2} = \beta I_B + \beta(\beta + 1) I_B \approx \beta^2 I_B$

$U_{BE} = U_{BE_1} + U_{BE_2} \approx 1,2\ V$

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GENERATOR CURENT CONSTANT

Dz - cond. inversă

$U_D = U_{Dz}$

$U_D = I_E R_2 + U_{EB} \Rightarrow I_E = \frac{U_D - U_{EB}}{R_2}$

$I_C \cong I_E = \frac{U_{Dz} - U_{EB}}{R_2}$

$I_D - I_1 - I_B > 0 \Rightarrow \frac{E - U_{Dz}}{R_1} - I_B > 0$

$I_B = \frac{I_C}{\beta}$

T. RAN : $U_{EC} > 0{,}6V$

$U_{EC} = E - I_E R_2 - U_{DL} > 0{,}6$

Ex.: $E = 12V$ $I_C = 10mA$ $U_{Dz} = 3V3$

$\Rightarrow R_2 = \frac{U_{Dz} - U_{EB}}{I_C} = 270\Omega$

$I_B = \frac{I_C}{\beta} > 0{,}1mA$ ALEGEM $I_1 = 1mA$

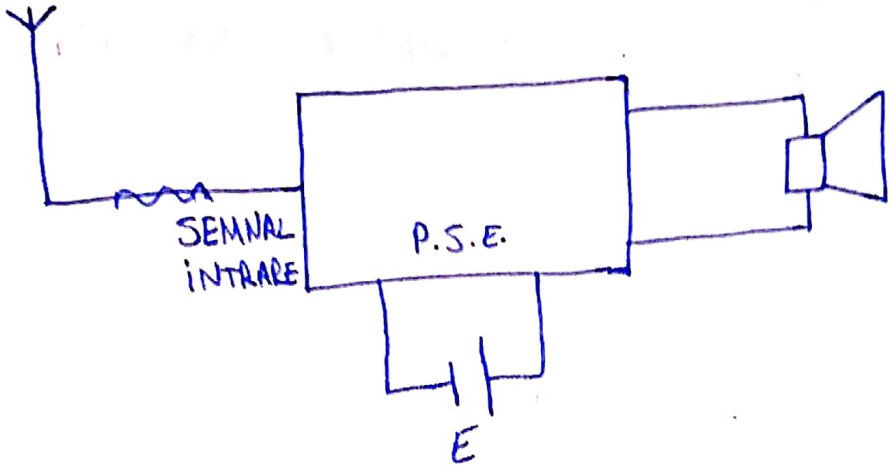
$R_1 = \frac{E - U_{Dz}}{I_1} = 8K7$

$U_{DL\,max} = E - R_2 I_C - U_{EC\,min} = 12V - 2{,}7V - 0{,}6V = 8{,}7V$

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15.XI.2022

Curs 10 - DEEA



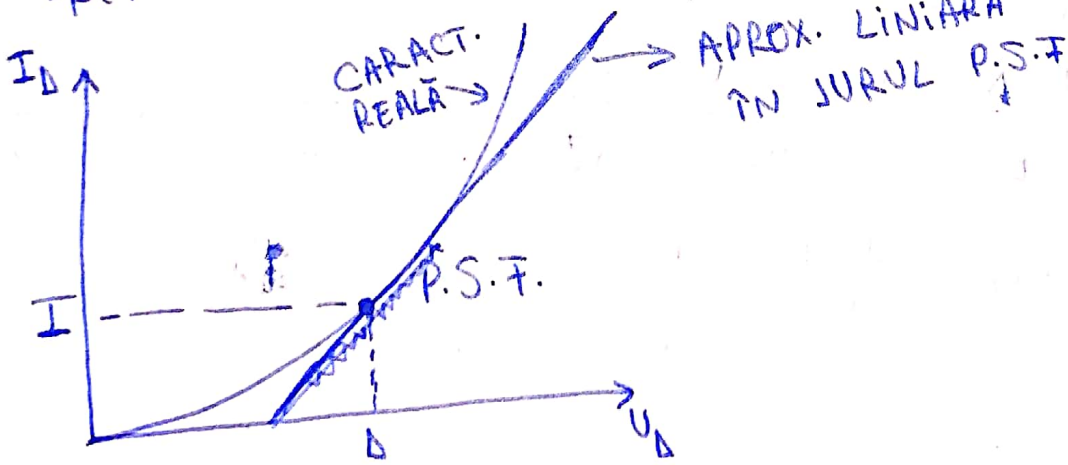
$$\begin{cases} u_i = A \sin \omega t \\ E \end{cases} \qquad u_0(t) = f(E, u_i(t)) \xrightarrow{\text{SUPERPOZIȚIE}} f(E,0) + f(0, u_i(t))$$

E → ALIMENTARE; $u_i(t)$ → SEMNAL UTIL

= dc. f liniară

$$f(\lambda a + \beta b) = \lambda f(a) + \beta f(b)$$

$$f(\lambda a) = \lambda f(a)$$



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REGIM DINAMIC

- CIRC. ELECTRONICE
- APROX. DE SEMNAL MIC

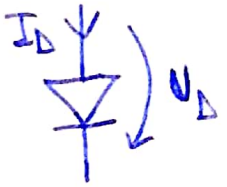
$u_i(t)$ - mic - aprox. lin. în jurul P.S.F. e suficient de precisă

$f(E, u_i)$

1) Calc. P.S.F. $f(E,0)$
2) Liniarizăm circuitul în jurul P.S.F.
3) Calc. $f'(0, u_i)$ — $f'(\ )$ - liniar } Aprox. semnal mic
4) Superpoziție - efectul sumei e suma efectelor

$$f(E, u_i) = f(E,0) + f'(0, u_i)$$

Aprox. semnal mic (reg. dinamic) diodă



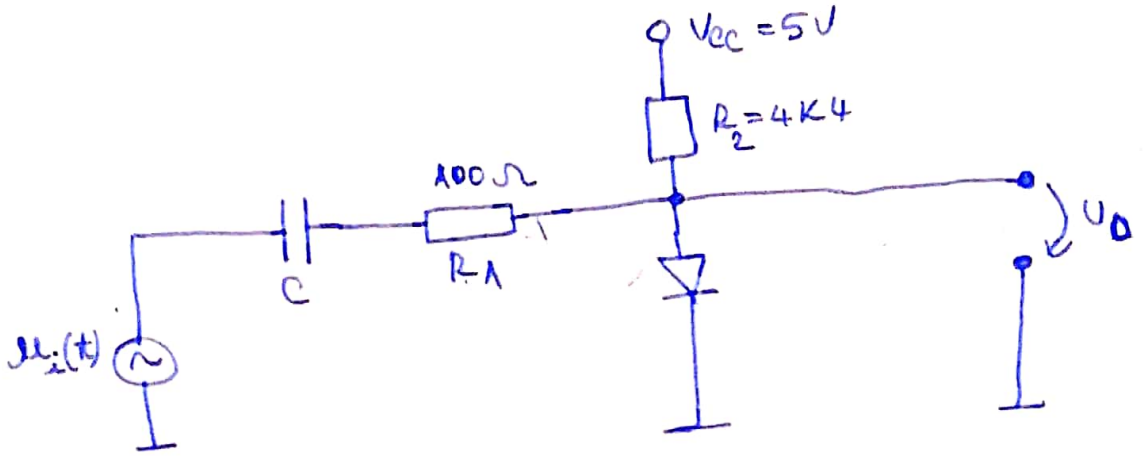
$$I_D = I_S\left(e^{\frac{U_D}{V_t}} - 1\right) \approx I_S e^{\frac{U_D}{V_t}}$$

$$\frac{dI_D}{dU_D} = \frac{d\, I_S e^{\frac{U_D}{V_t}}}{dU_D} = \frac{I_S}{V_t} e^{\frac{U_D}{V_t}} = \frac{I_D}{V_t} \text{ - conductanță}$$

$$R_{\text{DINAMICĂ DIODĂ}} = R_{DD} = \frac{dU_D}{dI_D} = \frac{V_t}{I_D} \approx \frac{25\,mV}{I_D} = \frac{1}{40 I_D} \; K\Omega$$

(I_D în mA)

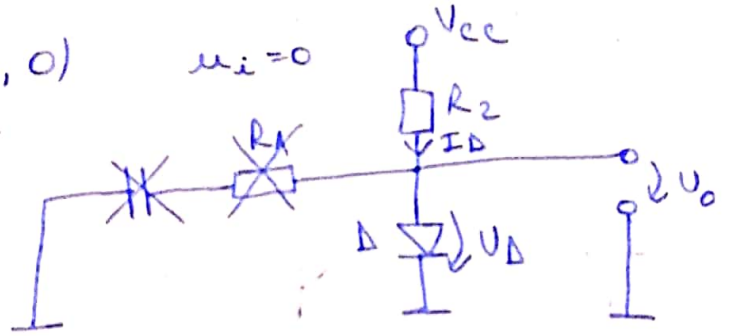
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1) P.S.F. $f(E,0)$ $\quad u_i=0$

V_{cc}

$I_c = C\dfrac{dV_c}{dt}$



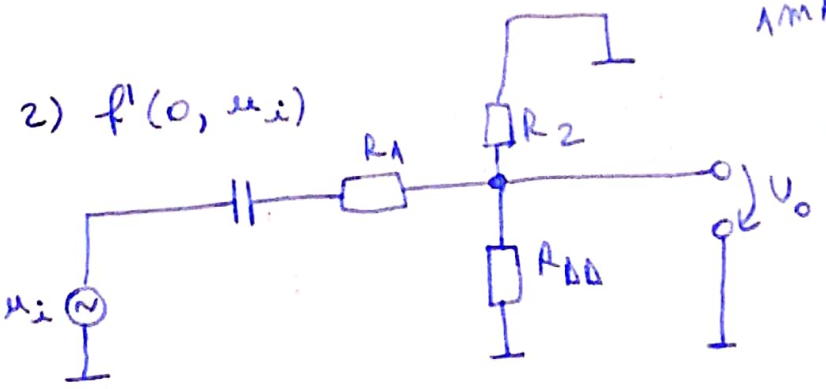
$V_{cc} = I_D R_2 + U_D \Rightarrow I_D = \dfrac{V_{cc}-U_D}{R_2} \simeq \dfrac{5V-0,6V}{4K4} = 1mA$

$U_o = U_D - 0,6V$ "$f(E,0)$"

Aproc. S.M. diodă $\quad R_{DD} = \dfrac{1}{40\, I_D} = 25\Omega$

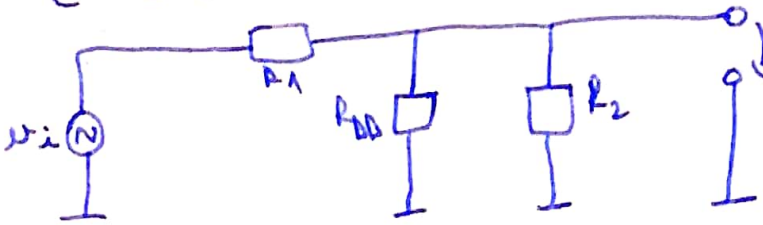
I_D = 1mA

2) $f'(0,u_i)$



$X_c = \dfrac{1}{j\omega C} \simeq 0$

C - SCURTCIRCUIT PT. u_i

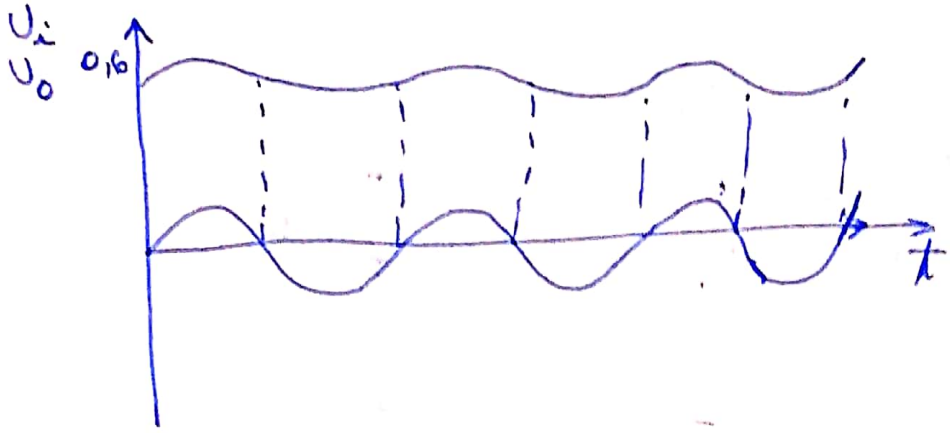


$u_o = u_i \dfrac{R_{DD} \parallel R_2}{R_{DD} \parallel R_2 + R_1}$

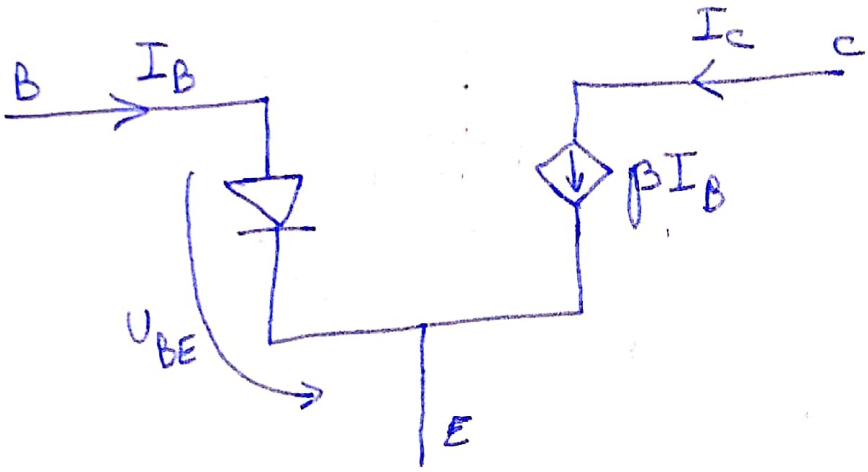
"$f'(0,u_i)$"

$U_o(0,u_i) = u_i \cdot \dfrac{25\Omega \parallel 4K4}{25\Omega \parallel 4K4 + 100\Omega} \simeq 0,2\, u_i$

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Tranzistoare model semnal mic

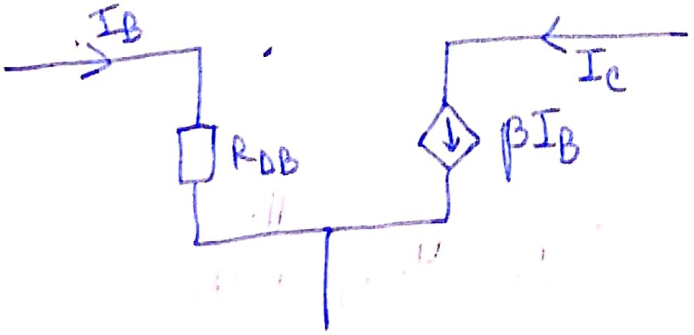


$$I_C = \beta I_B$$

$$I_C = I_S\left(e^{\frac{U_{BE}}{V_T}} - 1\right)$$

$$\frac{dI_B}{dU_{BE}} = \frac{I_B}{V_T} \qquad R_{DB} = \frac{dU_{BE}}{dI_B} = \frac{V_T}{I_B} = \frac{1}{40 I_B}\ K\Omega$$

(I_B in mA)



$$R_{BB'} = \frac{1}{40 I_B} = \frac{1}{40\frac{I_C}{\beta}} = \frac{\beta}{40 I_C}\ K\Omega$$

(I_C in mA)

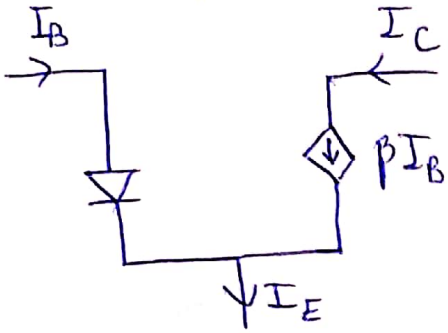
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17.XI.2022

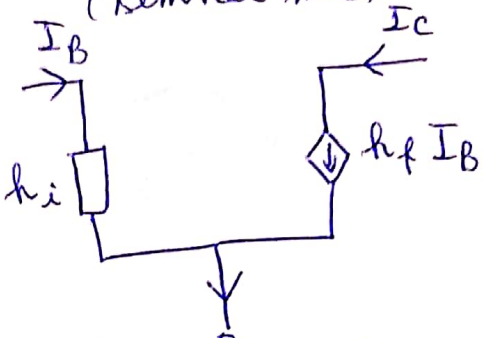
Curs 11 - DEEA

REGIM DINAMIC T. BIP

Reg. static

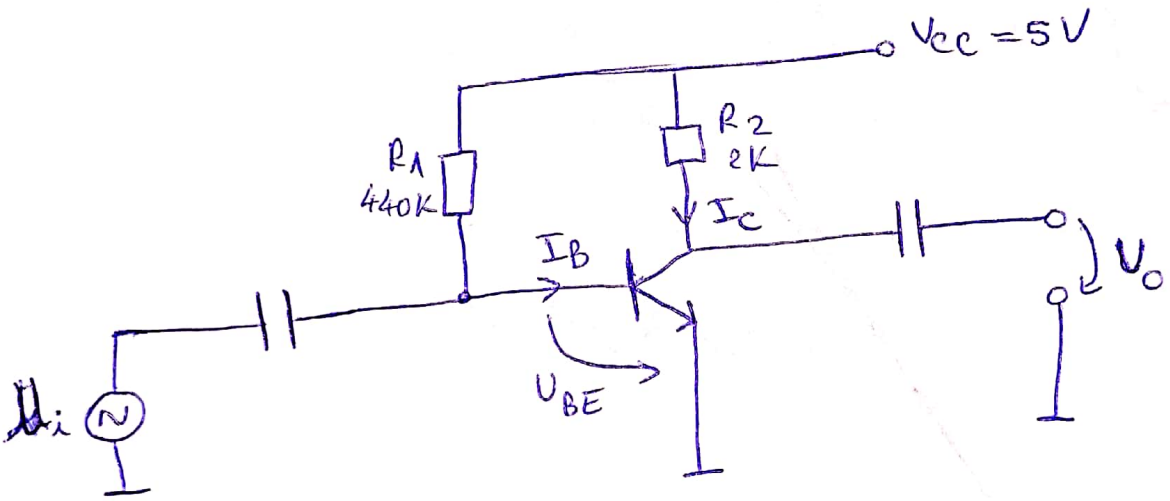


Reg. dinamic (semnal mic)



$$h_i = \frac{\beta}{40 I_{C_{mA}}} \ K$$

$$h_f = \beta$$



1) PST $u_i = 0$ $\Rightarrow$:

$V_{CC} = R_1 I_B + U_{BE} \Rightarrow I_B = 0{,}01\,mA \Rightarrow I_C = \beta I_B = 1\,mA$

$U_{CE} = V_{CC} - I_C R_2 = 3V > 0{,}6$ RAN

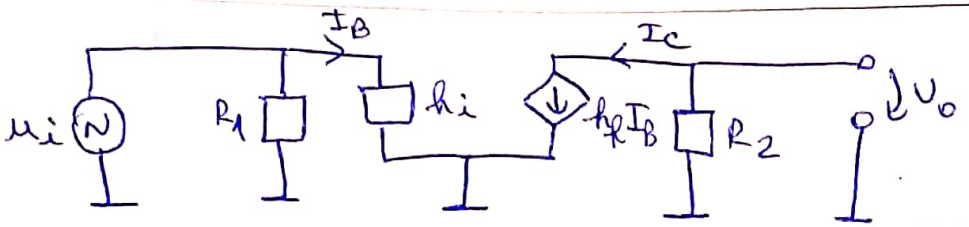
2) ~~Liniarizare~~ Liniarizare aprox. semnal mic

$h_i = \frac{100}{40 \cdot 1} = 2{,}5K$, $h_f = \beta = 100$

3) Pasivizam V_{CC} $\Rightarrow$ Calc. $U_o(u_i)$

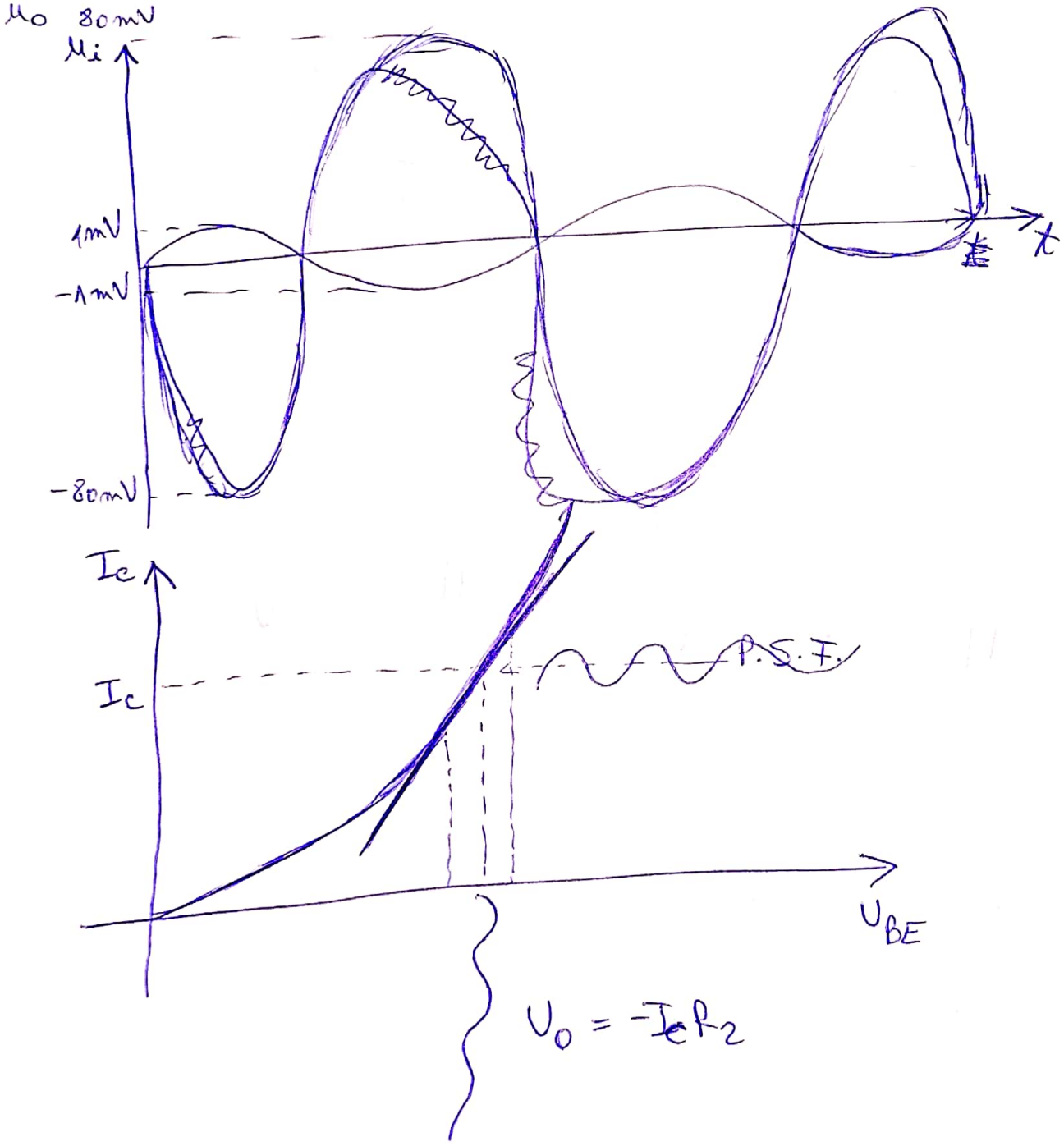
Regim dinamic

Scanned with CamScanner



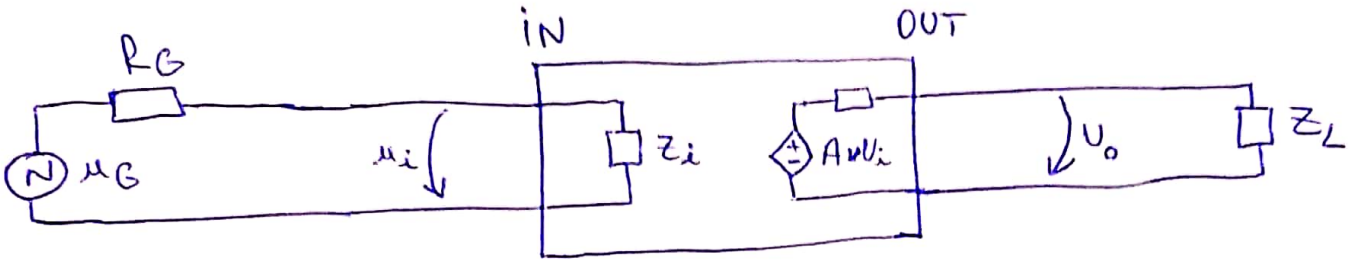
$$U_o = -I_C R_2 = -h_f I_B R_2 = -h_f \frac{u_i}{h_i} R_2 = -100 \cdot \frac{2K}{2{,}5K} \cdot u_i =$$

$$= -80\, u_i$$



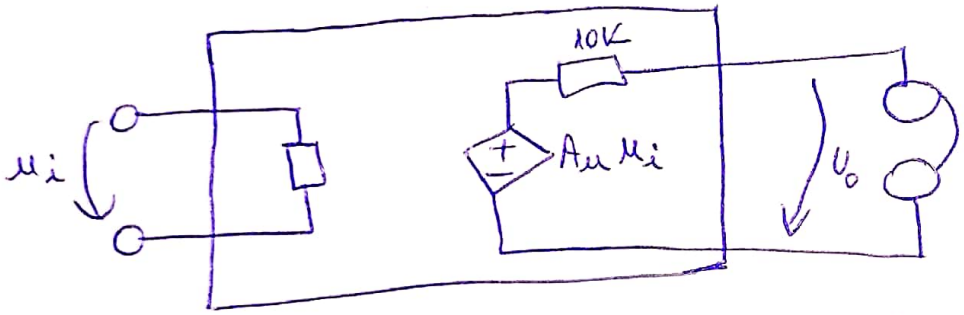
Scanned with CamScanner

Amplificatoare



$$U_o = A_u u_i \frac{z_L}{z_L + z_o} = A_u \frac{z_L}{z_L + z_o} \frac{z_i}{z_i + R_G} u_G$$

↑n mod ideal $z_i = \infty$, $z_o = 0$



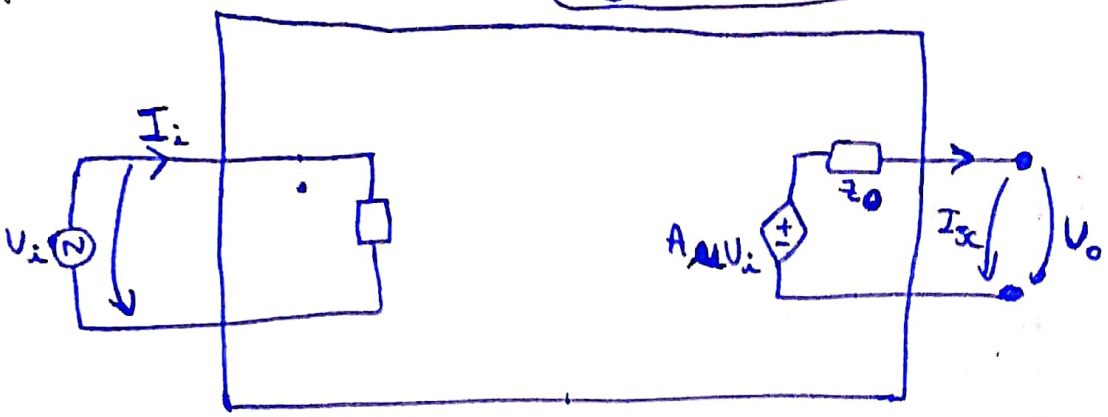
$z = 32\,\Omega$

$$U_o = \frac{z}{z + 10K} A_u u_i = \frac{32}{32 + 10K} A_u u_i \approx \frac{1}{300} A_u \cdot u_i$$

Scanned with CamScanner

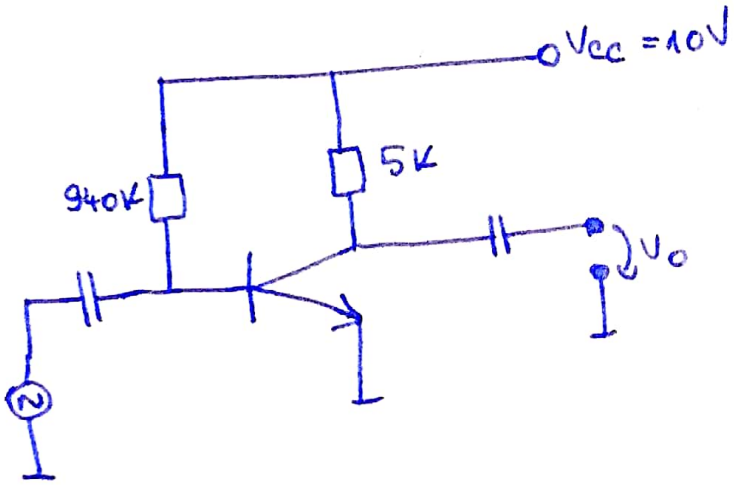
24.XI.2022

Curs 12 - DEEA



$$z_i = \frac{U_i}{I_i} \qquad I_{SC} = \frac{A_U U_i}{z_G}$$

$$A_0 = \frac{U_{GOL}}{U_i} \qquad z_0 = \frac{A_U U_i}{I_{SC}}$$

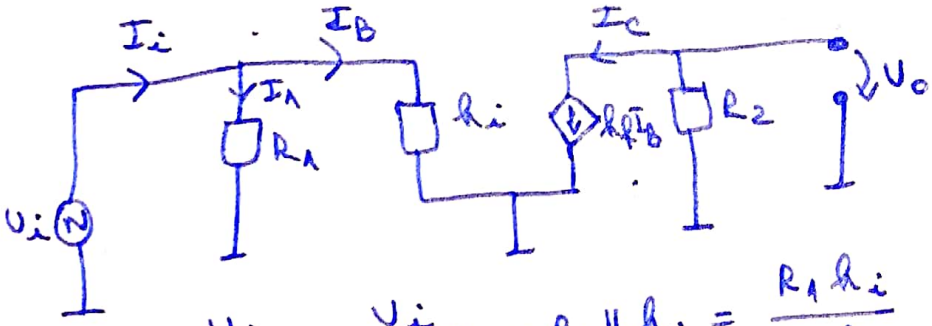


$$\frac{I_C}{\beta} R_1 + U_{BE} = V_{CC} \Rightarrow I_C = 1mA$$

$$U_{CE} = V_{CC} - R_2 I_C = 5V \geq 0,6$$ RAN

$$R_i = \frac{\beta}{40 I_C} \approx 2,5K$$

$$R_f = \beta = 100$$



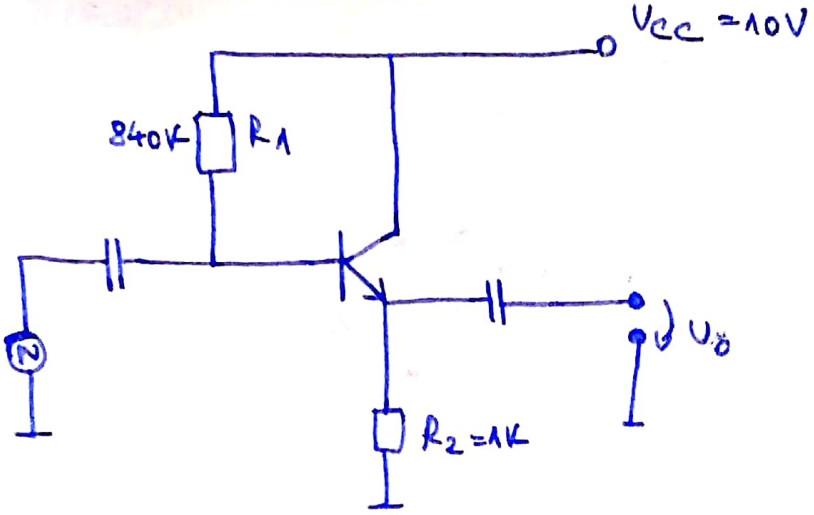
$$z_i = \frac{U_i}{I_i} = \frac{U_i}{\frac{U_i}{R_1} + \frac{U_i}{R_i}} = R_1 \| R_i = \frac{R_1 R_i}{R_1 + R_i}$$

$$A_{OL} = \frac{U_{OGOL}}{U_i} = \frac{-I_C R_2}{U_i} = \frac{\frac{U_i}{R_i} \cdot R_f R_2}{U_i} = -\frac{R_f}{R_i} R_2 = \quad (R_f = \beta)$$

$$= -\frac{100 \cdot 5K}{2,5K} = -200$$

$$z_0 = \frac{I_{SC}}{A_U \cdot U_i} \frac{A_U \cdot U_i}{I_{SC}} = \frac{-\frac{R_f}{R_i} R_2 R_i}{-\frac{U_i}{R_i} R_f} = R_2 = 5K$$

Scanned with CamScanner

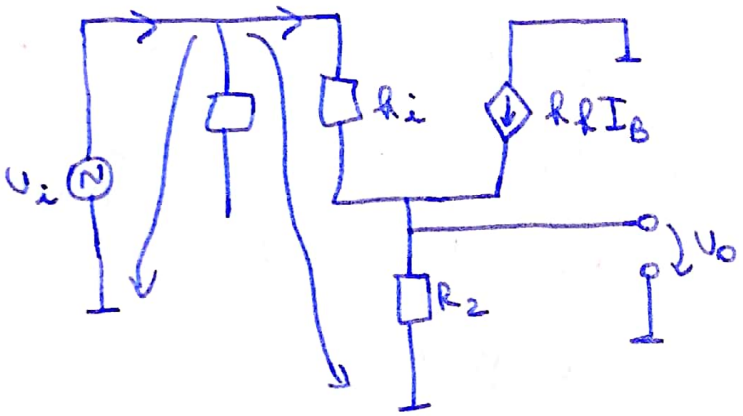


PSF $\quad I_B R_1 + V_{BE} + I_E R_2 = V_{CC}$

$I = 1mA$

$V_{CE} = 9V$

$h_i = \frac{\beta}{40 I_C} = 2{,}5K \qquad h_f = \beta = 100$



$$V_{o\,GOL} = I_E R_2 - (h_f + 1) I_B R_2$$

$$V_i = \frac{I_B h_i}{V_{h_i}} + \frac{R_2 I_B (h_f+1)}{V_{R_2}}$$

$$I_B = \frac{V_i}{h_i + R_2(h_f+1)} \approx \frac{V_i}{R_2 h_f}$$

$$V_{o\,GOL} = (h_f+1)\frac{V_i}{R_2 h_f} R_2 \approx V_i$$

$$A_V = \frac{V_{GOL}}{V_i} \approx 1$$

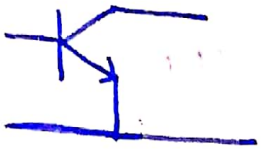
$$z_i = \frac{V_i}{I_i} = \frac{V_i}{\frac{V_i}{R_1} + \frac{V_i}{R_2 h_f}} = R_1 \| R_2 h_f \approx 90K$$

$$z_o = \frac{V_{o\,GOL}}{I_{SC}} = \frac{V_i}{I_B(1+h_f)} = \frac{V_i}{\frac{V_i}{R_f}(1+h_f)} = \frac{h_i}{1+h_f} \approx \frac{h_i}{h_f} \approx 25\Omega$$

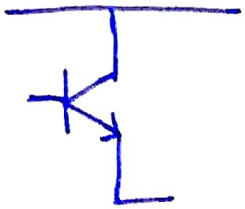
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EC (Emitor comun) Av Zi Zo

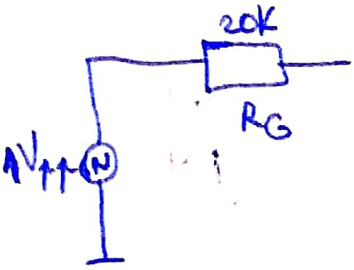
EM -200 2,5K 5K

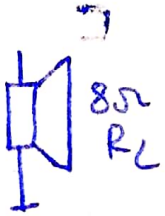


CC

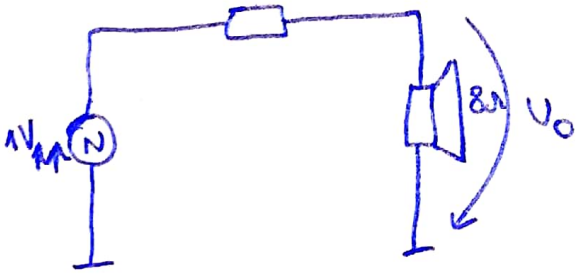
CM  1 90K 25Ω

LINE OUT

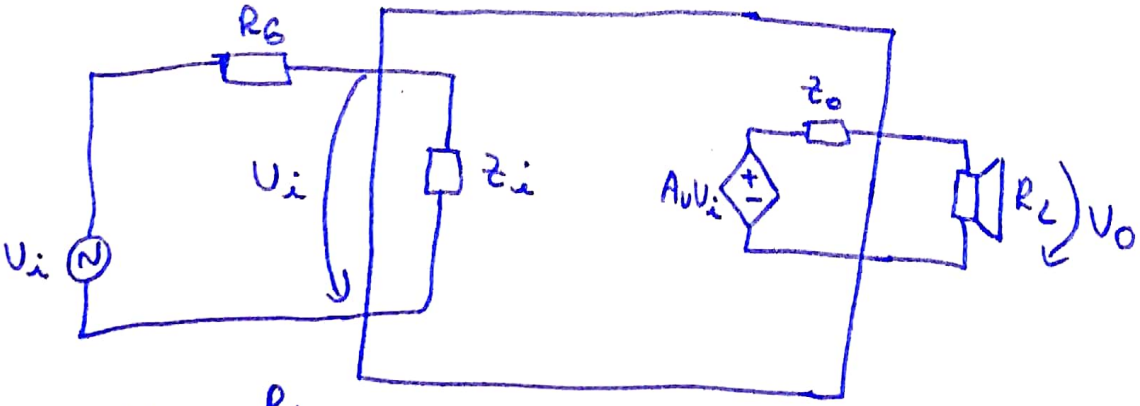




1) Conexiune directă
2) Emitor comun (EC)
3) Colector comun (CC)



$$U_o = U_i \cdot \frac{R_L}{R_G + R_L} = 1V_{\uparrow\uparrow} \cdot \frac{8\Omega}{8\Omega + 20000\Omega} = 0,0004 V_{\uparrow\uparrow} = 0,4mV_{\uparrow\uparrow}$$



$$U_o = \frac{R_L}{z_o + R_L} \cdot \frac{z_i}{R_G + z_i} A_U \cdot U_i$$

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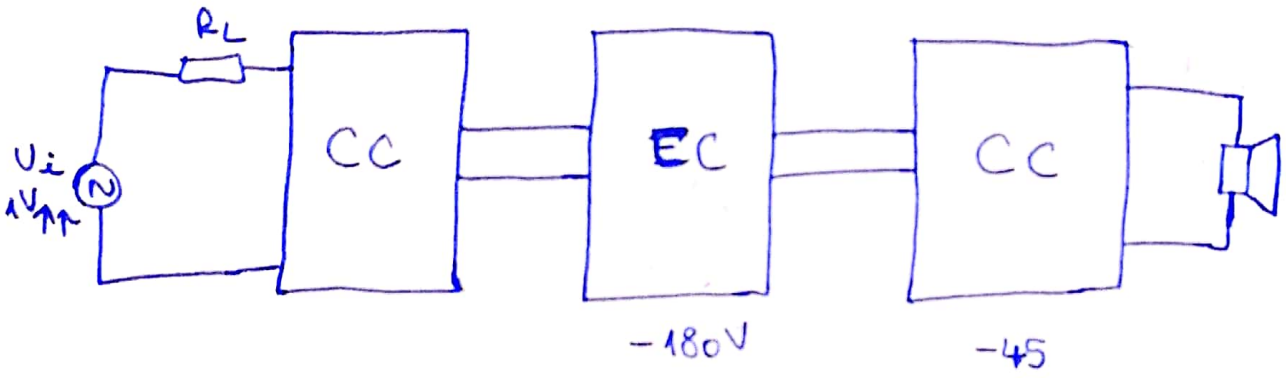
2) EC $U_o = \frac{8\Omega}{4K+8\Omega} \cdot \frac{2,5K}{20K+2,5K} \cdot (-200) \cdot 1V_{\uparrow\uparrow} = -0,035\,V_{\uparrow\uparrow}$

3) CC $U_o = \frac{8\Omega}{25\Omega+8\Omega} \cdot \frac{90K}{90K+20K} \cdot 1 \cdot 1V_{\uparrow\uparrow} = 0,2\,V_{\uparrow\uparrow}$

1) CD $U_o = 0,0004\,V_{\uparrow\uparrow}$

EC $U_o = -0,035\,V_{\uparrow\uparrow}$

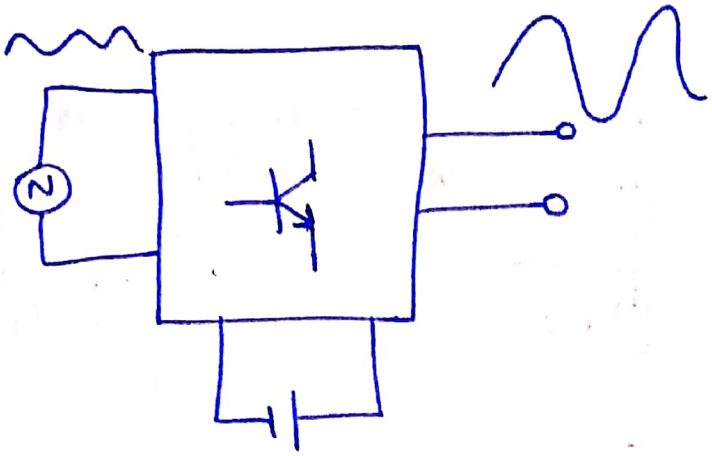
CC $U_o = 0,2\,V_{\uparrow\uparrow}$ $P = \frac{U^2}{R}$

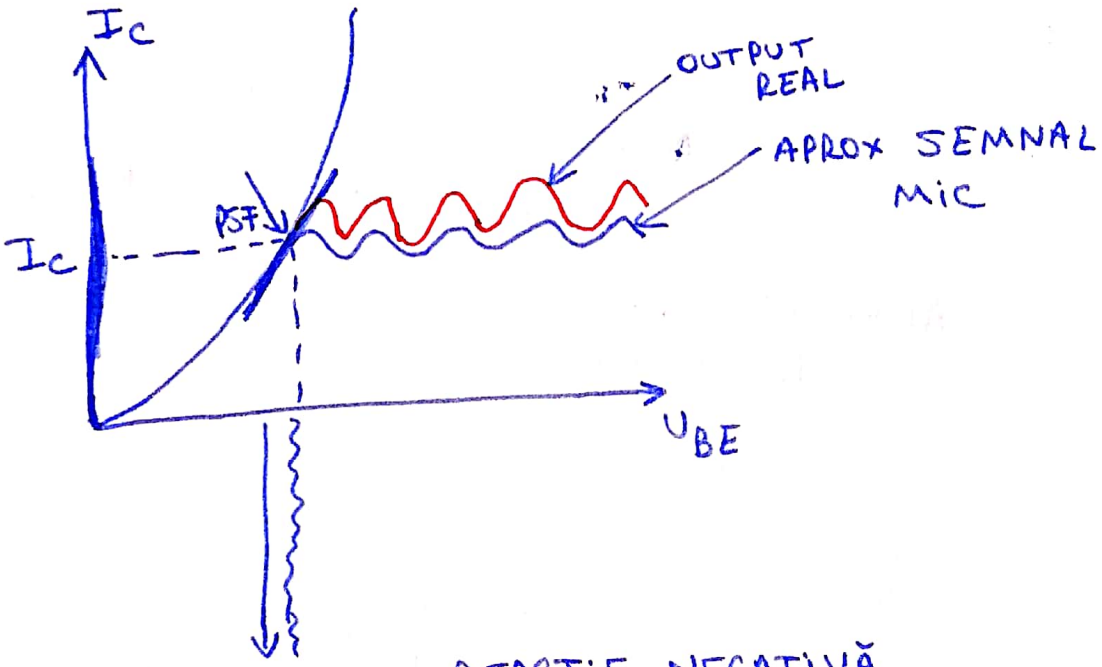


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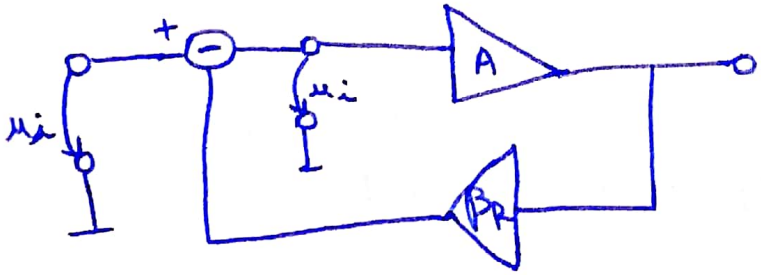
Curs 13 - DEEA





REACȚIE NEGATIVĂ

REACȚIE → inputul se duce la output



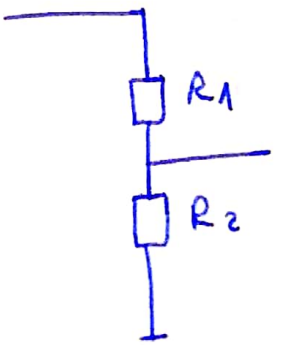
$$U_o = A U_{iA}$$
$$U_{iA} = U_i - U_o \beta_R \Rightarrow U_o = A(U_i - \beta_R U_o)$$
$$U_o = U_i \frac{A}{1 + \beta_R A}$$

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$$A_R = \frac{A}{1+\beta_R A}$$

$$\lim_{A\to\infty} A_R = \lim_{A\to\infty} \frac{A}{1+\beta_R A} = \frac{1}{\beta_R}$$





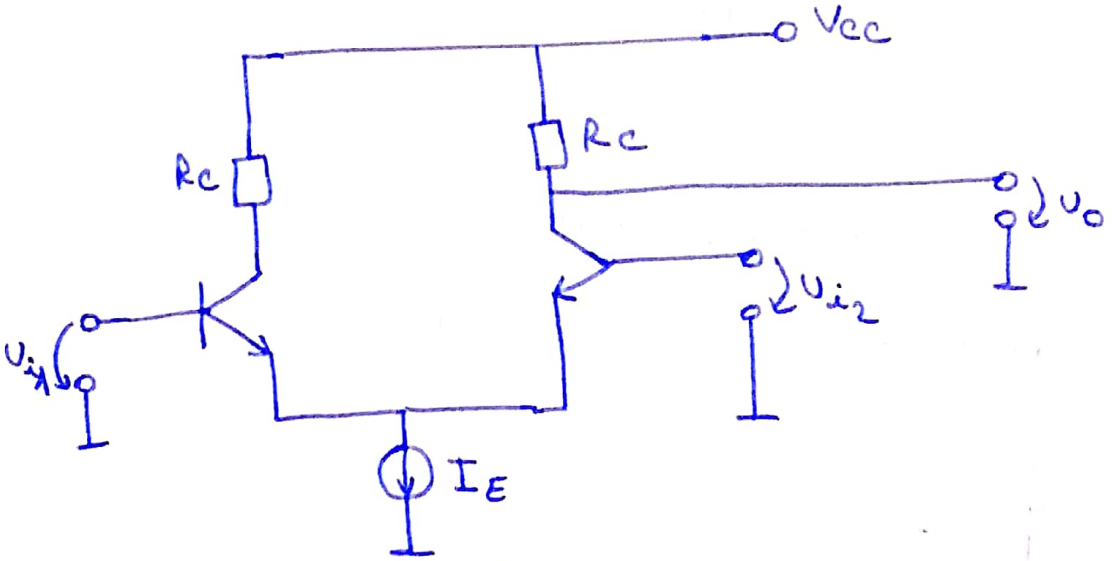
divisor de tensiune
↳ input mai mare decât output

$$\beta_R = \frac{R_2}{R_1+R_2}$$

$$\lim_{A\to\infty} \frac{d A_R}{d A} = 0$$

A → cu mai multe tranzistoare
↳ input mai mic decât output

AMPLIFICATOARE DIFERENȚIALE



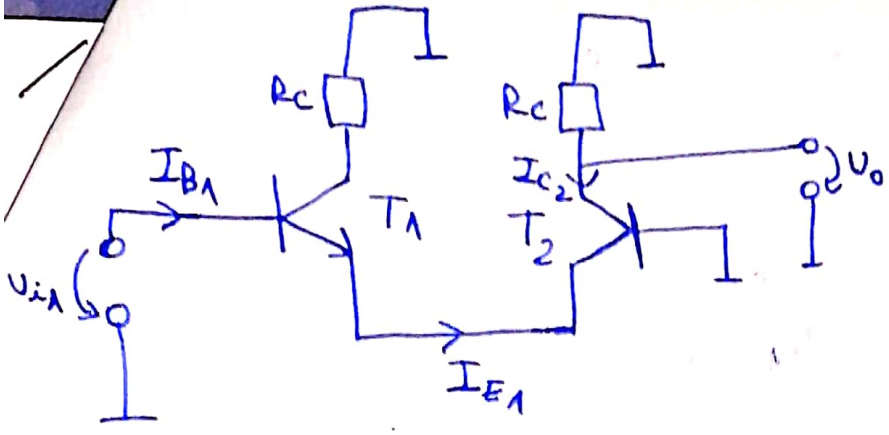
$$V_o(V_{i1}, V_{i2}) \overset{\text{APROX LINIARĂ}}{=\!=} V_o(V_{i1}, 0) + V_o(0, V_{i2})$$

SUPER POZIȚIE

$$V_o(V_{i1}, 0) = ?$$

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Polarizat

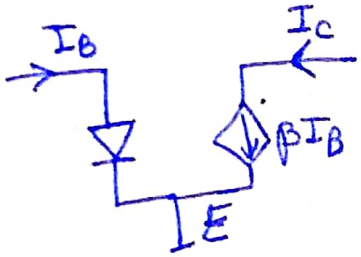


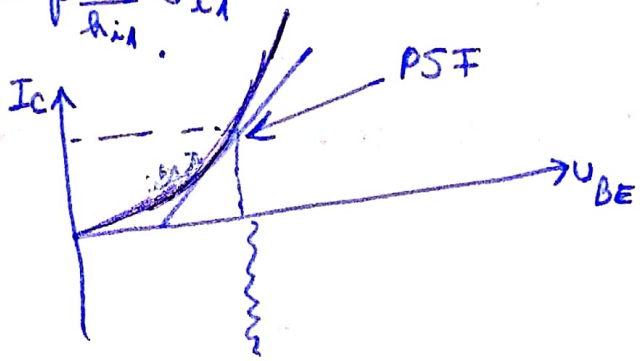
$$U_{i1} = I_{B1} h_{i1}$$

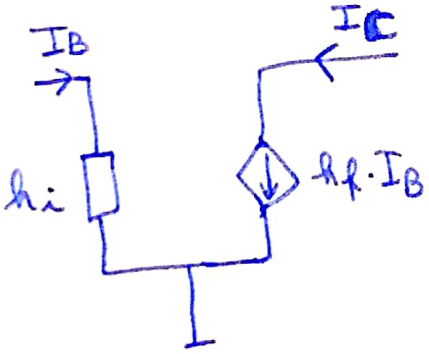
$$I_{E1} \cong \beta I_{B1}$$

$$I_{C2} \cong I_{E2} = -I_{E1}$$

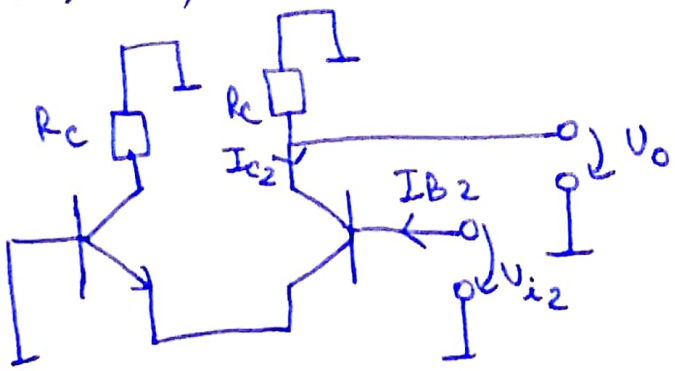
$$U_0 = -I_2 R_c = I_{E1} R_c = \beta \frac{R_c}{h_{i1}} U_{i1}$$







$U_0(0, U_{i2})$



$$I_{B2} = \frac{U_{i2}}{h_{i2}} ; \quad I_{C2} = \beta I_{B2}$$

$$U_0 = -I_{C2} \cdot R_C = -R_C \beta \cdot \frac{U_{i2}}{h_{i2}}$$

$$U_0 = \beta \frac{R_C}{h_{i1}} U_{i1} - \beta \frac{R_C}{h_{i2}} U_{i2}$$

$$h_i = \frac{\beta}{I_C} ??$$

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T_1 identic cu T_2 și polarizați identic

$h_{i1} = h_{i2} = h_i$

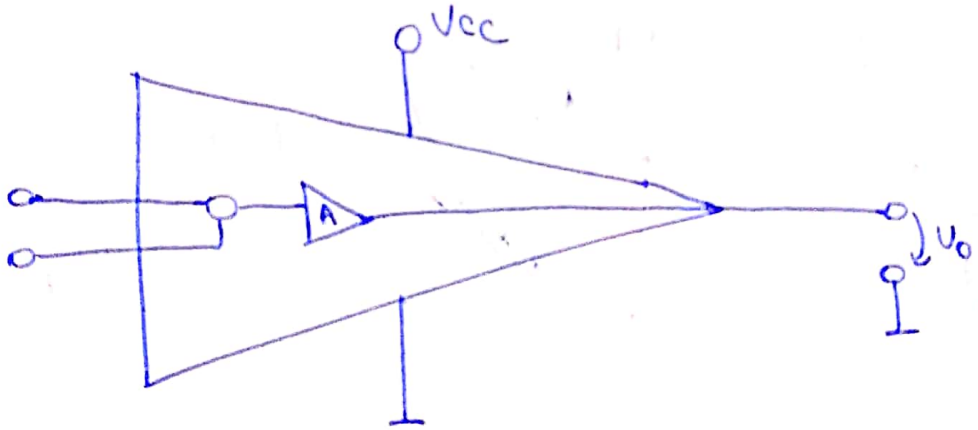
$U_o = \beta \frac{R_C}{h_i} (U_{i1} - U_{i2})$ ← MOD COMUN ??

$U_o = A_D (U_{i1} - U_{i2}) + A_C (U_{i1} + U_{i2})$

dorim A_D - mare

A_C - 0 sau cât mai mică

AMPLIFICATOR OPERAȚIONAL



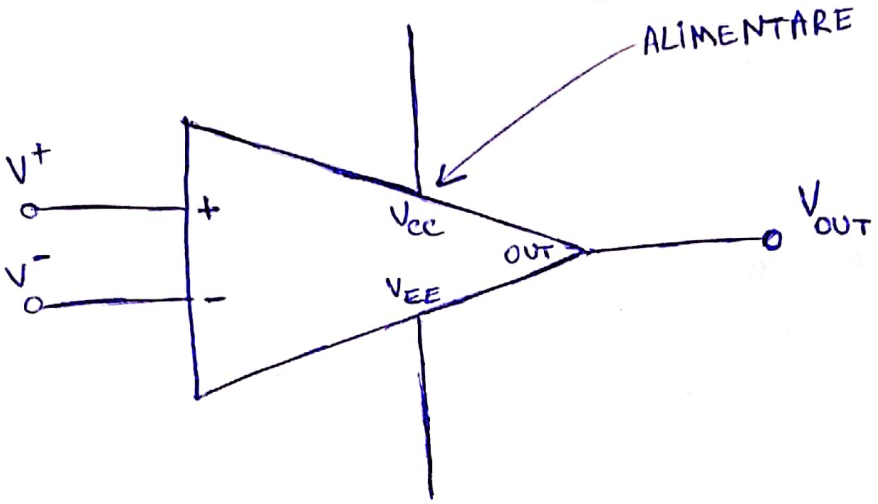
$U_o = A (V^+ - V^-)$

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8. XII . 2022

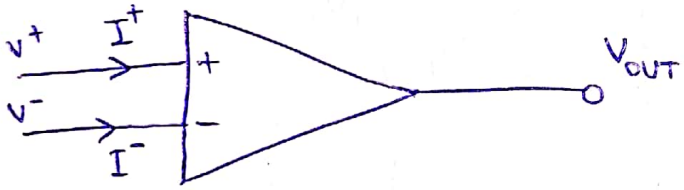
Curs 14 - DEEA

AMPLIFICATOARE OPERAȚIONALE



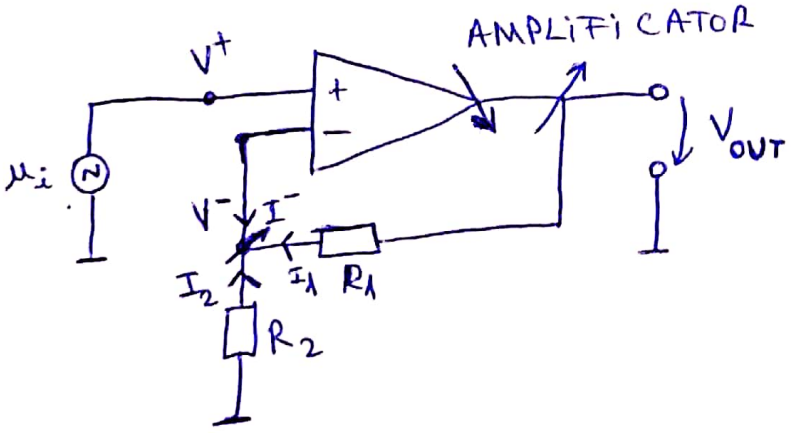
$$V_{OUT} = A(V^+ - V^-)$$

MODEL A.O. IDEAL



$$\begin{cases} V_{OUT} = A(V^+ - V^-) \\ A \to \infty \quad (A = \infty) \\ I^+ = I^- = 0 \quad z_i = \infty \end{cases}$$

AMPLIFICATOR NEINVERSOR



$$V_{OUT} = A(V^+ - V^-)$$

$$V^+ = u_i$$

$$I^- + I_2 + I_1 = 0$$

$$0 + \frac{0 - V^-}{R_2} + \frac{V_{OUT} - V^-}{R_1} = 0$$

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$$V^-\left(\frac{1}{R_2}+\frac{1}{R_1}\right)=\frac{V_{OUT}}{R_1}$$

$$V^- = \frac{V_{OUT}}{R_1}\cdot\frac{R_1R_2}{R_1+R_2} = V_{OUT}\cdot\frac{R_2}{R_1+R_2}$$

$$V_{OUT} = A\left(\mu_i - V_{OUT}\frac{R_2}{R_1+R_2}\right)$$

$$V_{OUT} = \frac{A\mu_i}{1+A\frac{R_2}{R_1+R_2}} \overset{A\to\infty}{=\!=} \mu_i\frac{R_1+R_2}{R_2}$$

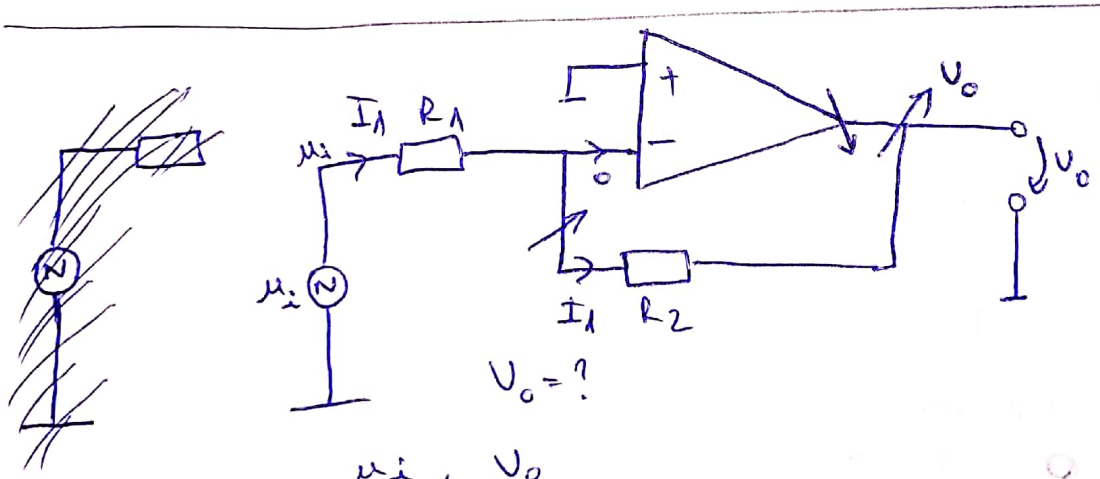
Ex.: DORIM $A=20$

$R_1=19K$ $\quad R_1=190K$

$R_2=1K$ $\quad R_2=10K$

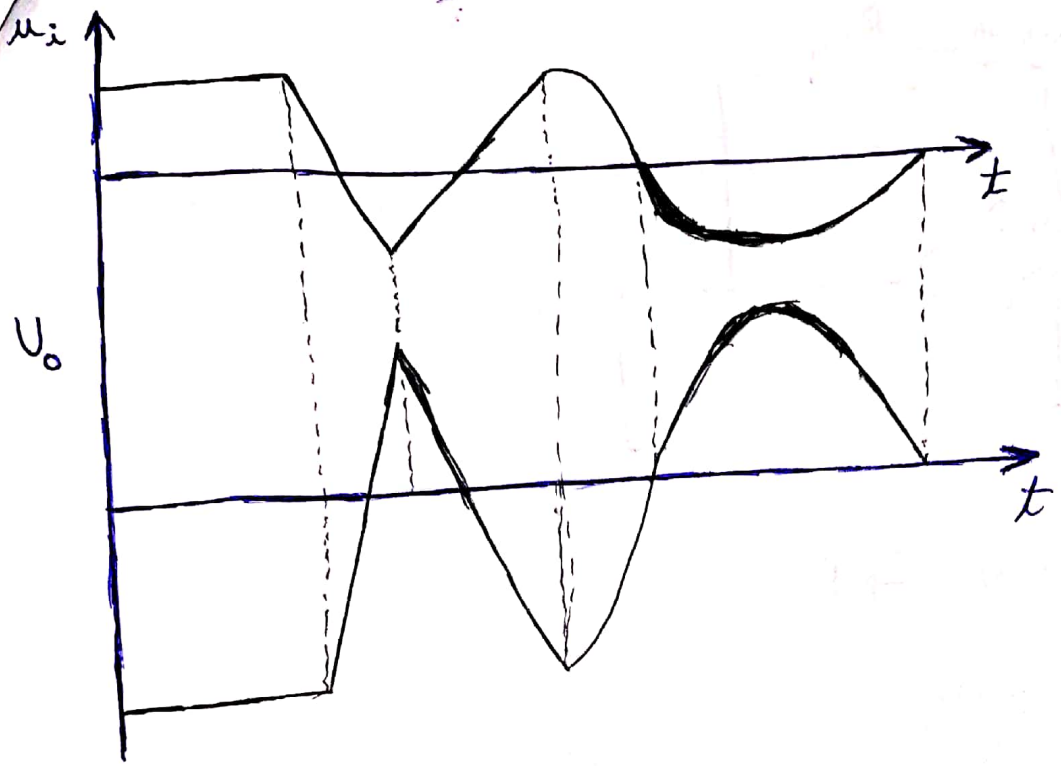
$$V^+-V^- = \frac{V_{OUT}}{A} = \frac{\mu_i}{A}\frac{R_1+R_2}{R_2} \overset{A\to\infty}{=\!=} 0$$

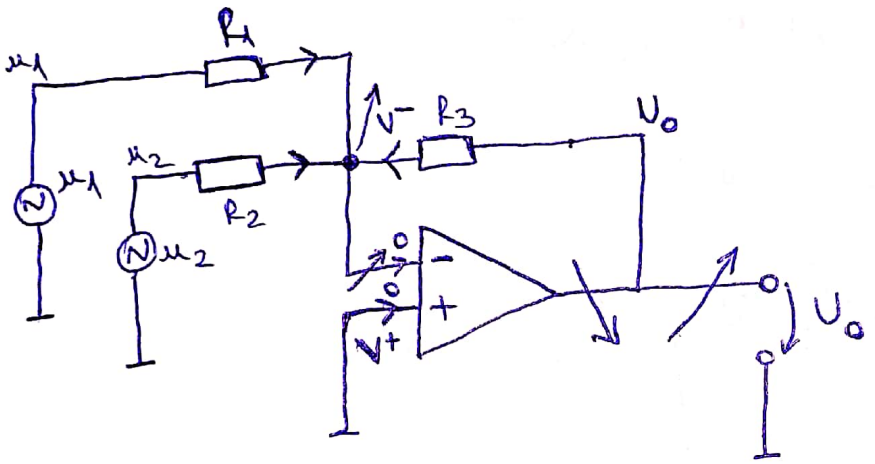
R.N. $\Rightarrow V^+=V^-$



$$V_{OUT} = -A\cdot\frac{\frac{\mu_i}{R_1}+\frac{V_o}{R_2}}{\frac{1}{R_1}+\frac{1}{R_2}}$$

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R.N. $\Rightarrow V^+ = V^- = 0$

$$\frac{u_1 - V^-}{R_1} + \frac{u_2 - V^-}{R_2} + \frac{U_o - V^-}{R_3} = 0$$

$$\frac{u_1}{R_1} + \frac{u_2}{R_2} = -\frac{U_o}{R_3}$$

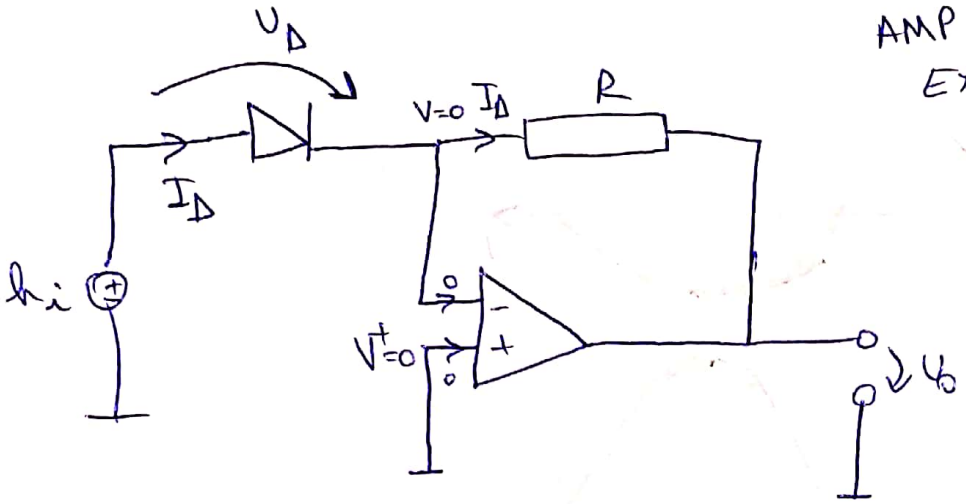
$$U_o = -R_3\left(\frac{U_1}{R_1} + \frac{U_2}{R_2}\right)$$

ALEGEM $R_1 = R_2 = R_3$

$U_o = -(U_1 + U_2)$

Scanned with CamScanner

AMP EXPONENTIAL

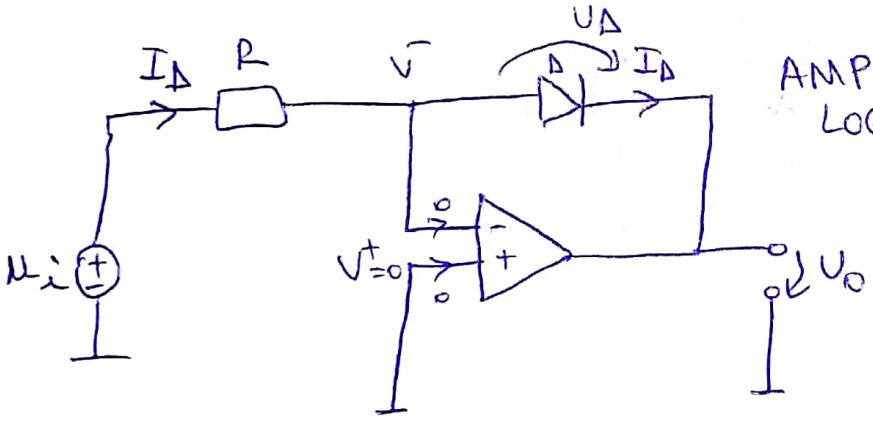


$$\frac{V^- - U_o}{R} = I_\Delta \Rightarrow U_o = I_\Delta R$$

$$V^- - V^- = U_\Delta \Rightarrow U_\Delta = U_i$$

$$I_\Delta = I_S\left(e^{\frac{U_\Delta}{V_T}} - 1\right) \approx I_S e^{\frac{U_\Delta}{V_T}}$$

$$U_o = R I_S e^{\frac{U_\Delta}{V_T}} \approx \exp(U_\Delta)$$



AMP LOGARITMIC

$$V^+ = 0 \quad R.N. \Rightarrow V = V^+ = 0$$

$$I_\Delta = \frac{u_i - V^-}{R} = \frac{u_i}{R}$$

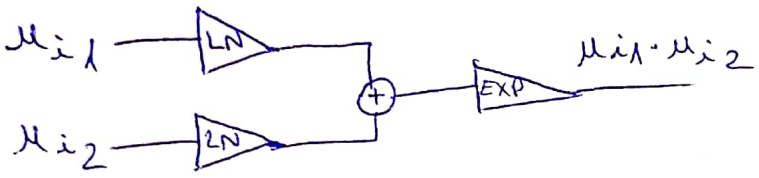
$$\left.\begin{aligned} I_\Delta &= I_S e^{\frac{U_\Delta}{V_T}} \\ U_\Delta &= V^- - U_o = -U_o \end{aligned}\right\} \Rightarrow I_S e^{-\frac{U_o}{V_T}}$$

$$-\frac{U_o}{V_T} = \ln\frac{I_\Delta}{I_S} \Rightarrow U_o = -V_T \ln\frac{I_\Delta}{I_S} = -V_T \ln\frac{u_i}{I_S R} \approx \ln u_i$$

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$$u_{i1} \cdot u_{i2} = \exp(\ln u_{i1} + \ln u_{i2})$$

+ −
ln exp

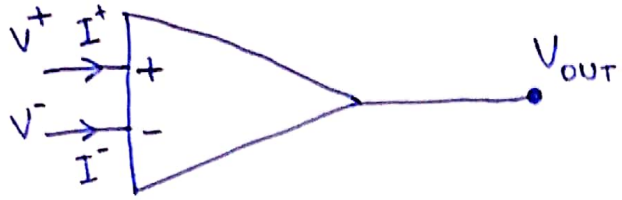


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Curs 15
DEEA

A.O.



$V_{OUT} = A(V^+ - V^-)$

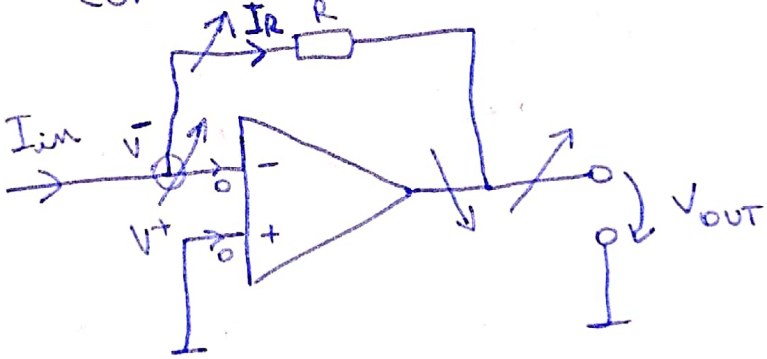
$A \simeq 10^6 \simeq \infty$

$I^+, I^- \to 0$
$pA \to mA$

R.N. $\Rightarrow$ LINIAR $V^+ \approx V^-$

A INV
A Neinversor
Sumator
A LOG
A EXP

CONV. CURENT → TENSIUNE



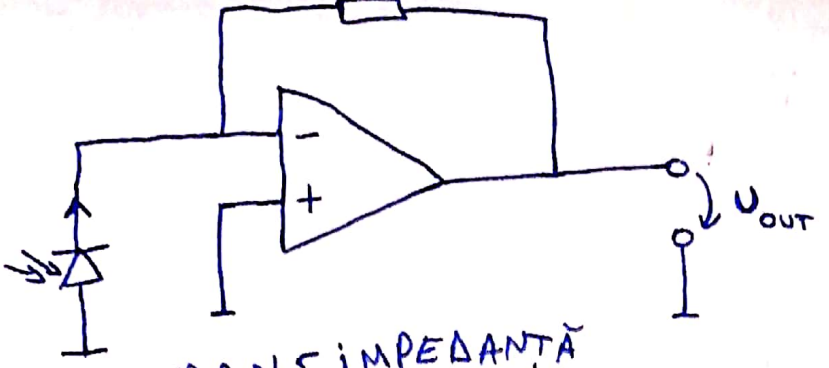
R.N. $\Rightarrow V^+ = V^-$ $\}V^- = 0$
$V^+ = 0$

KCL: $I_{in} = I_R = \frac{V^- - V_{OUT}}{R} = -\frac{V_{OUT}}{R}$

$V_{OUT} = -RI_R$ (ca să fie fără − punem un invertor)

!!! Subiect de examen !!!

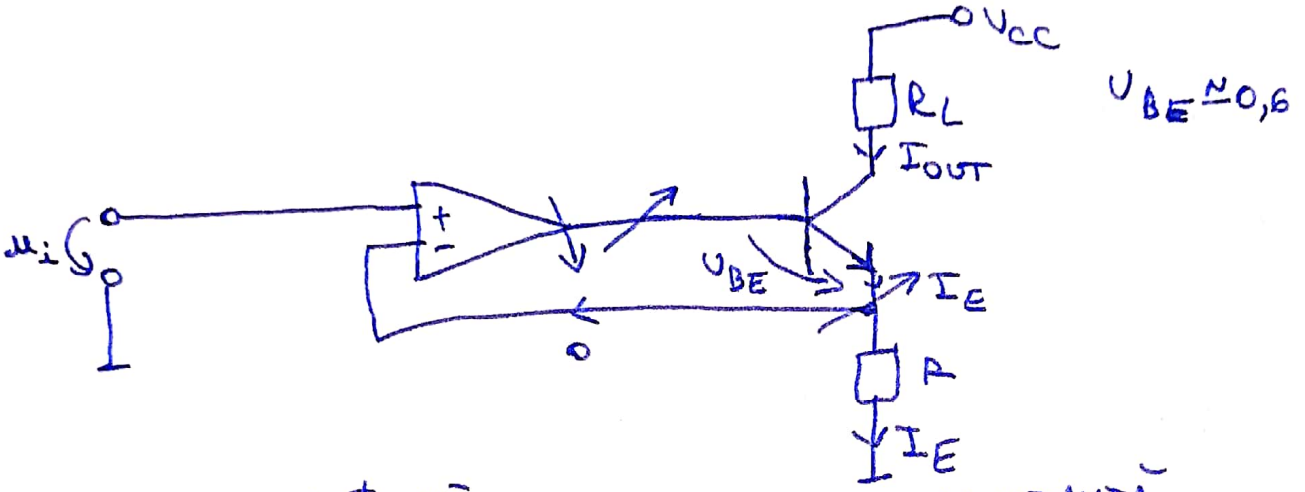
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A/D

TRANSIMPEDANȚĂ

CONVERTOR TENS. → CURENT



$U_{BE} \cong 0,6$

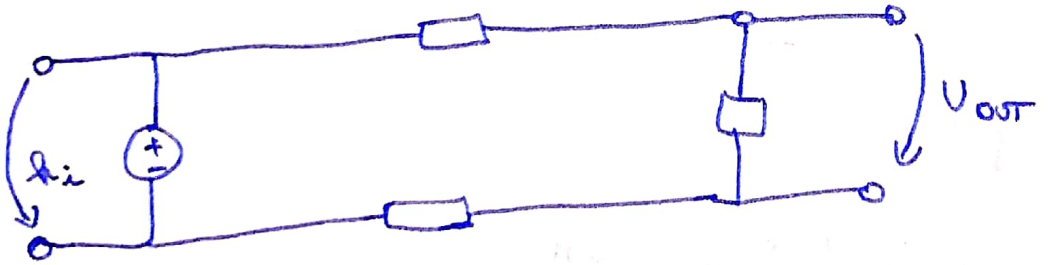
R.N. $V^+ = V^-$

$V^+ = u_i$

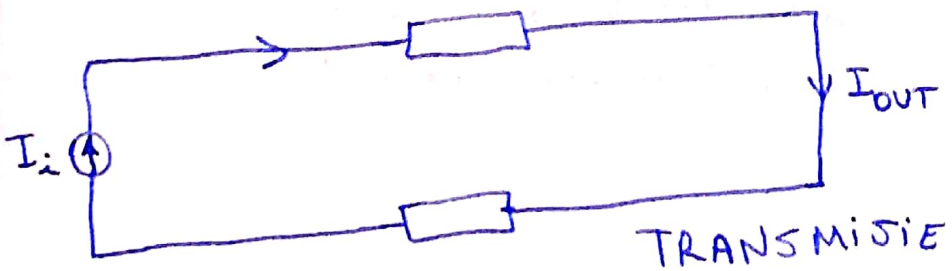
$V^- = RI_E = u_i$

$I_{OUT} = I_C \cong I_E = \frac{1}{R} u_i$

TRANSCONDUCTANȚĂ



SENZOR



TRANSMISIE ÎN CURENT

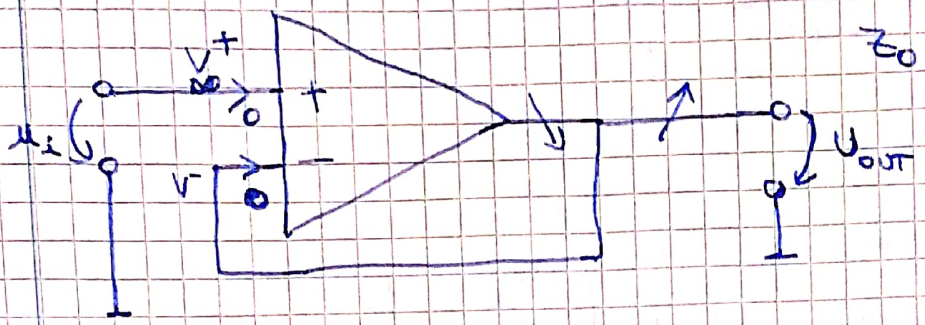
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REPETOR TENSIUNE

$U_0 = U_i$

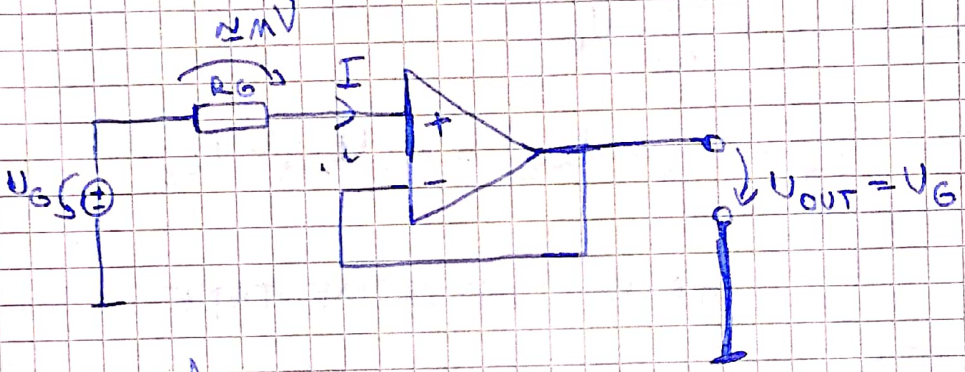
$Z_i = \infty$

$Z_0 \approx 0$



R.N. $\Rightarrow V^+ = V^-$

$V^+ = u_i \quad V^- = U_0 \quad \Rightarrow U_0 = u_i$

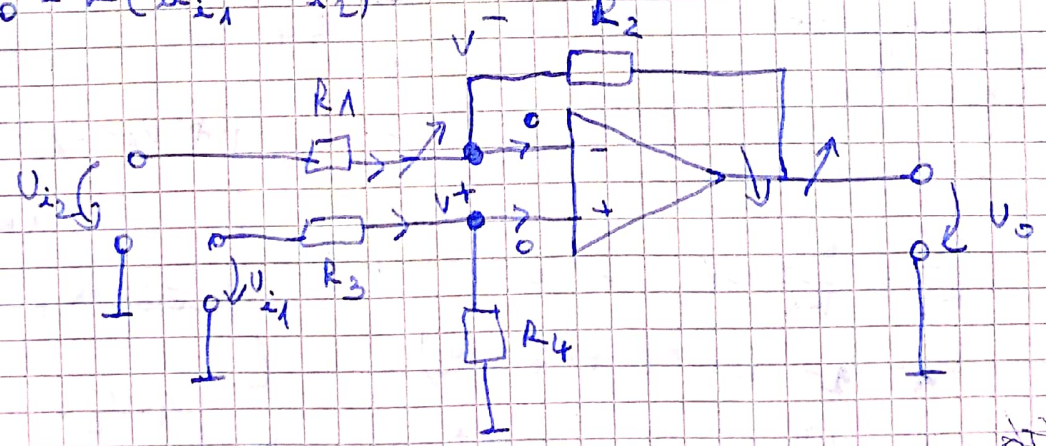


I $\approx$ nA

$R_G \approx$ 1K

AMPLIFICATOR DIFERENȚIAL

$U_0 = K(u_{i1} - u_{i2})$



R.N. $\Rightarrow V^+ = V^-$

$\frac{U_{i2} - V^-}{R_1} = \frac{U_0 - V^-}{R_2}$ = 0 $\qquad \frac{U_{i1} - V^+}{R_3} + \frac{0 - V^+}{R_4} = 0$

$U_0 = U_{i1} - U_{i2}$

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$$V^-\left(\frac{1}{R_1}+\frac{1}{R_2}\right)=\frac{U_{i2}}{R_1}+\frac{U_0}{R_2}$$

$$V^+\left(\frac{1}{R_3}+\frac{1}{R_4}\right)=\frac{U_{i1}}{R_3}$$

$$V^+=V^-$$

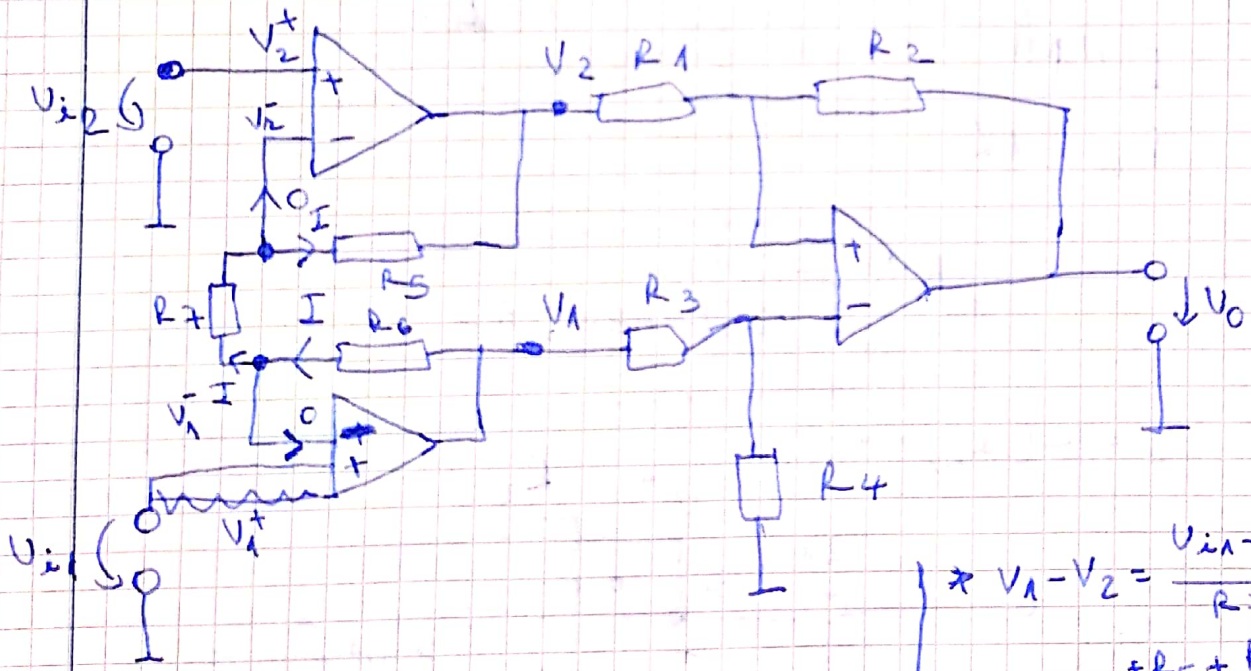
$$\frac{U_{i1}}{R_3}\cdot\frac{R_3R_4}{R_3+R_4}=\frac{R_1R_2}{R_1+R_2}\left(\frac{U_{i2}}{R_1}+\frac{U_0}{R_2}\right)$$

$$\frac{R_4}{R_3}=\frac{R_2}{R_1}=K \Rightarrow \frac{R_4}{R_3+R_4}=\frac{R_2}{R_1+R_2}$$

$$U_{i1}=U_{i2}+\frac{R_1}{R_2}U_0$$

$$U_0=\frac{R_2}{R_1}(U_{i1}-U_{i2})=K(U_{i1}-U_{i2})$$

AMPL. INSTRUMENTAȚIE



$$\frac{R_4}{R_3}=\frac{R_2}{R_1}=K \Rightarrow U_0=K(V_1-V_2)$$

$$V_1-V_2=I(R_6+R_7+R_5)$$

R.N. $V_1^+=V_1^-=U_{i1}$ $\quad V_2^+=V_2^-=U_{i2}$

$$U_{i1}-U_{i2}=IR_7 \quad *$$

$$* \; V_1-V_2=\frac{U_{i1}-U_{i2}}{R_7}(R_7+R_5+R_6)$$

ALEGEM $R_5=R_6=R$

$$V_1-V_2=(U_{i1}-U_{i2})\cdot\frac{R_7+2R}{R_7}$$

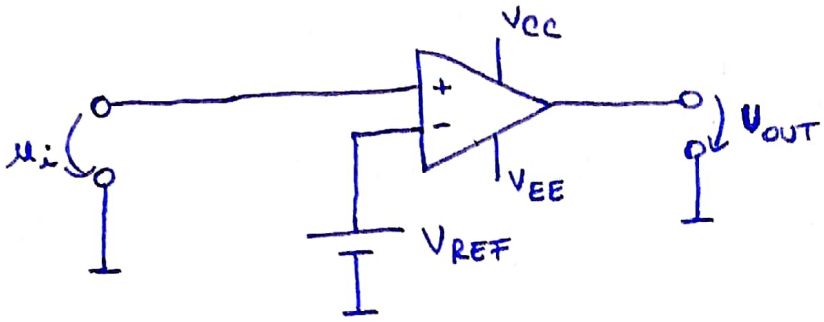
$$U_0=K\cdot\frac{R_7+2R}{R_7}(U_{i1}-U_{i2})$$

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Curs 16
Deea

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COMPARATOR

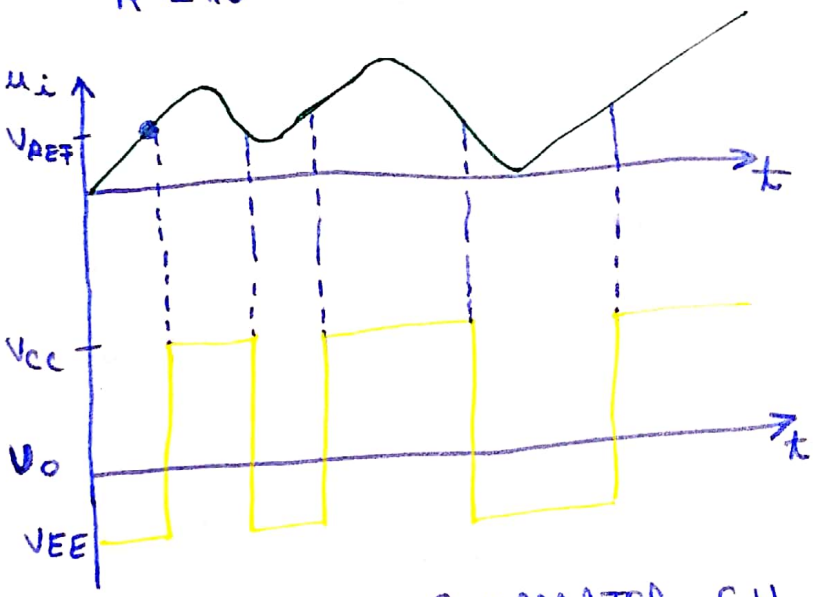


NU ARE R.N.

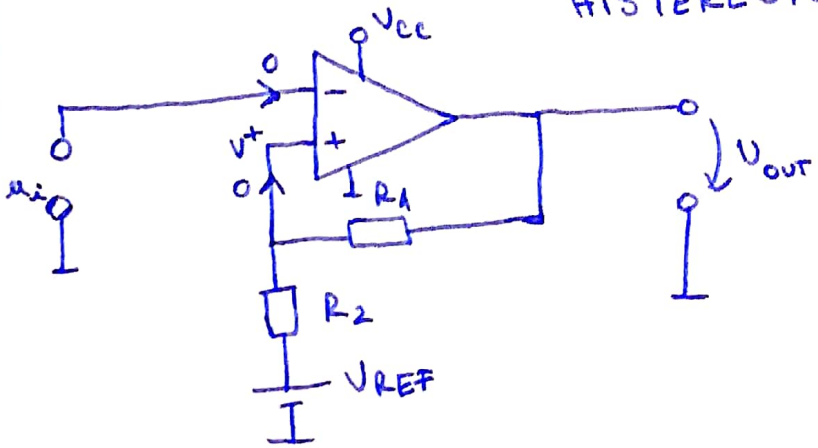
$U_{OUT} = A(V^+ - V^-) = A(u_i - V_{REF})$

$V_{EE} < U_{OUT} < V_{CC}$

$A \cong 10^6$



COMPARATOR CU HISTEREZIS



NU ARE R.N. ARE R.P.

ARE 2 STĂRI { $V_{OUT} = V_{CC}$ / $V_{OUT} = 0$

$V_{EE} = 0V$

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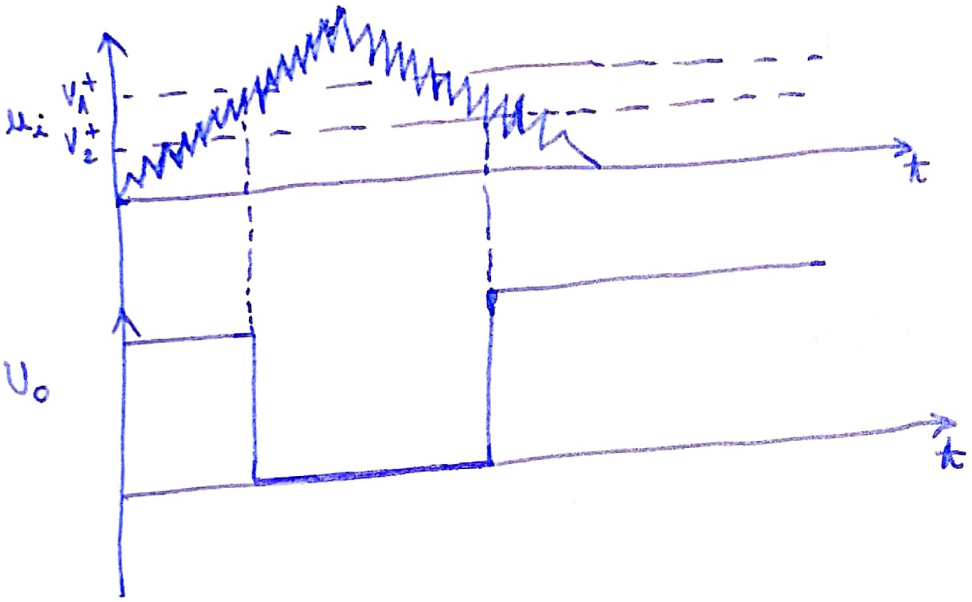
1) $V_{OUT} = V_{CC}$

$$V^+ = V_{REF} + V_{R_2} = V_{REF} + R_2 \cdot \frac{V_{OUT} - V_{REF}}{R_1 + R_2} =$$

$$= V_{REF} + R_2 \frac{V_{CC} - V_{REF}}{R_1 + R_2} = V_{REF} \frac{R_1}{R_1 + R_2} + V_{CC} \frac{R_2}{R_1 + R_2}$$

2) $V_{OUT} = 0$

$$V^+ = V_{REF} - R_2 \frac{V_{REF}}{R_1 + R_2} = V_{REF} \frac{R_1}{R_1 + R_2}$$



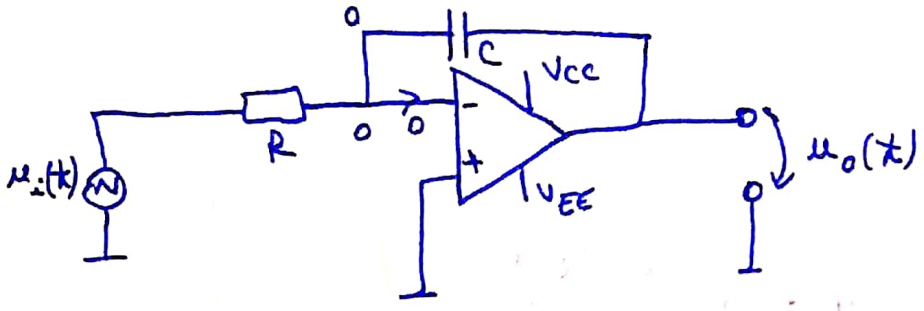
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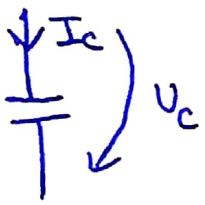
10.X.2023

Curs DEEA 17

Circuit integrator

$$u_0(t) = \int u_i(t)\,dt$$





$$I_c = C\frac{dV_c}{dt}$$

$$R.M. \Rightarrow \left.\begin{matrix} v^+ = v^- \\ v^+ = 0 \end{matrix}\right\} \Rightarrow v^- = 0$$

$$\frac{u_i(t) - 0}{R} + I_c = 0$$

$$U_c(t) = U_0(t) - 0 = U_0(t)$$

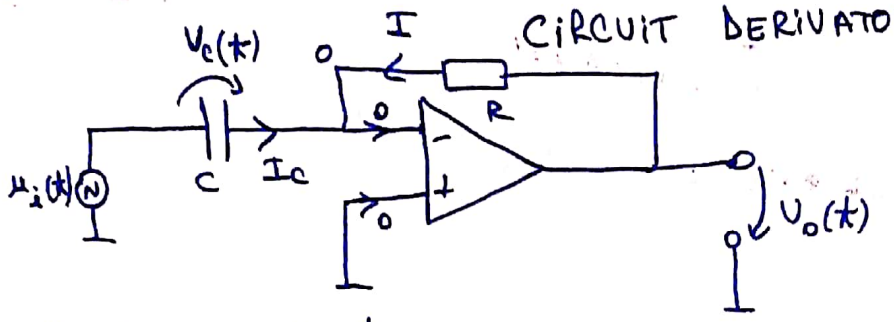
$$I_c = C\frac{dU}{dt} = C\frac{dU_0(t)}{dt}$$

$$\frac{u_i(t)}{R} + C\frac{dU_0(t)}{dt} = 0$$

$$\frac{dU_0(t)}{dt} = -\frac{1}{RC}U_i(t) \Rightarrow U_0(t) = -\frac{1}{RC}\int u_i(t)\,dt$$

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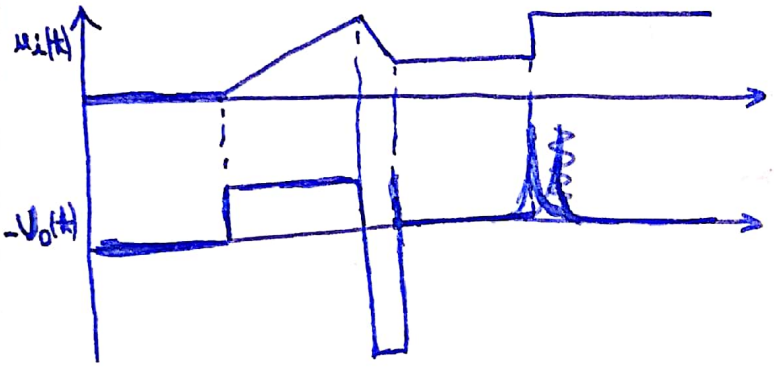
CIRCUIT DERIVATOR



R.N. $\Rightarrow \left.\begin{matrix} V^- = V^+ \\ V^+ = 0 \end{matrix}\right\} \Rightarrow V^- = 0$

$$C\frac{dU_c(t)}{dt} + \frac{U_o(t)-0}{R} = 0$$

$$C\frac{d(U_i(t)-0)}{dt} + \frac{U_o(t)}{R} = 0 \Rightarrow U_o(t) = -RC\frac{dU_i(t)}{dt}$$



FILTRE

- CIRC. CU AMPLIFICARE DEPENDENTĂ DE FRECVENȚĂ
- ANALIZĂ ÎN FRECVENȚĂ A UNUI CIRCUIT

CIRCUITE LINIARE INVARIANTE ÎN TIMP

$\mathcal{R}\{u_i(t)\} = u_o(t)$ $\quad u_i(t) \rightarrow \boxed{} \rightarrow U_o(t)$

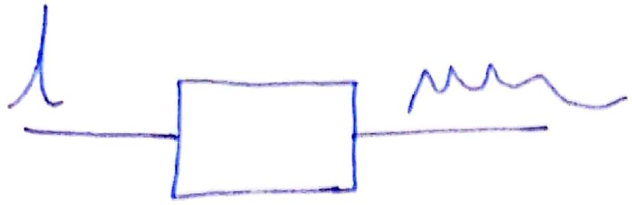
LINIAR: $\mathcal{R}\{\lambda u_i(t)\} = \lambda u_o(t)$

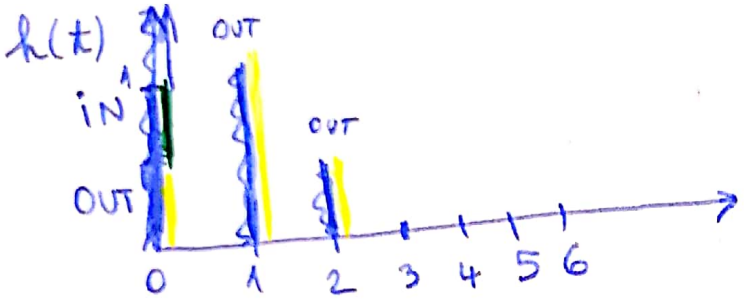
INV. ÎN TIMP: $\mathcal{R}\{u_i(t-\tau)\} = u_o(t-\tau)$

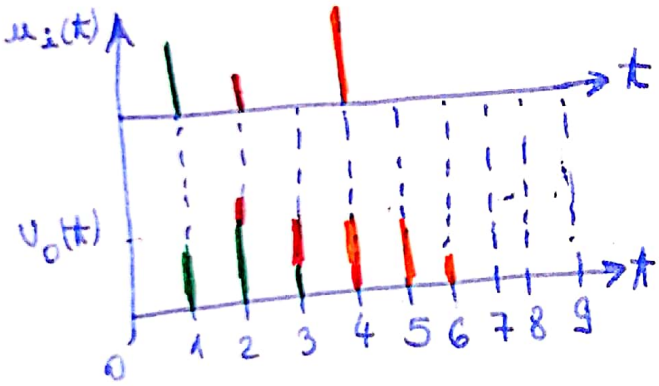
τ, λ const.

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L.T.i.







$u_i(t) \rightarrow \boxed{h(t)} \rightarrow V_o(t)$

$V_o(0) = V_i(0)\, h(0)$

$V_o(1) = V_i(0)\, h(1) + V_i(1)\, h(0)$

$V_o(2) = V_i(0)\, h(2) + V_i(1)\, h(1) + V_i(2)\, h(0)$

$$V_o(t) = \sum_{\tau=0}^{t} h(\tau)\, u_i(t-\tau)$$

$$u_i(t) * h(t) = \int_0^{\infty} h(\tau)\, h_i(t-\tau)\, d\tau$$

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$$e^{jwt} = \cos(wt) + j\sin(wt)$$

$$\cos(wt) = \frac{e^{jwt} + e^{-jwt}}{2}$$

$$u_i(t) = e^{jwt}$$

$h(t)$

$u_D(t) = ?$

$$u_D(t) = u_i(t) * h(t) = \int_0^\infty h(\tau) e^{j\omega(t-\tau)} d\tau =$$

$$= \int_0^\infty h(\tau) e^{jwt} e^{-jw\tau} d\tau =$$

$$= e^{jwt} \int_0^\infty h(\tau) e^{-jw\tau} d\tau =$$

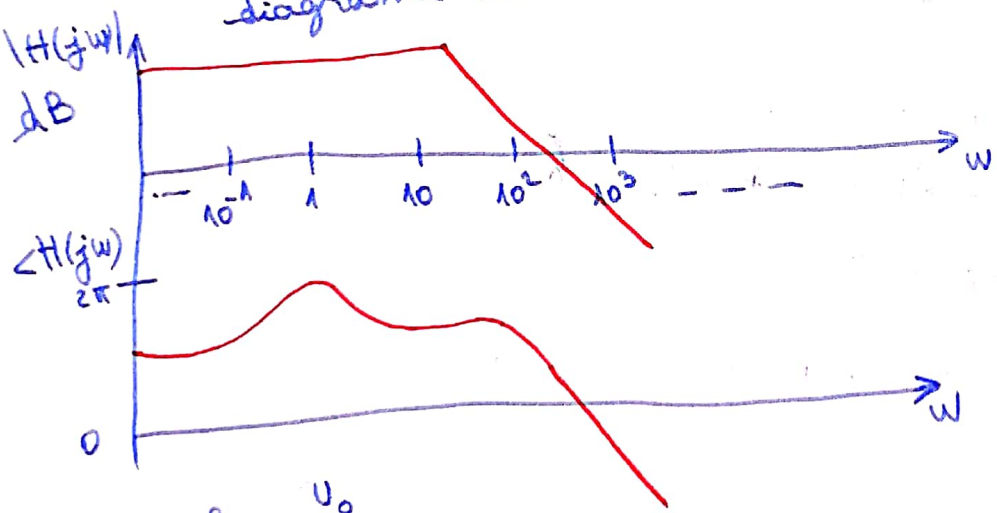
$$= H(jw) e^{jwt} = H(jw)\, u_i(t)$$

$$H(s) = \mathcal{L}\{h(t)\}$$

$$H(jw) = |H(jw)|\, e^{j \arg H(jw)}$$

$$u_o(t) = |H(jw)|\, e^{j \angle H(jw)} e^{jwt} = |H(jw)|\, e^{j(wt + \angle H(jw))}$$

diagrama BODE



$$dB = 20 \log_{10} \frac{U_o}{U_i}$$

$$u_i(t) = \sum_{i=0}^{n} A_i e^{j w_i t}$$

$\downarrow$ $h(t)$ $H(s)$

$$U_o(t) = \sum_{i=0}^{n} |H(jw_i)|\, A_i\, e^{j(w_i t + \angle H(jw_i))}$$

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Curs DEEA 18

12.I.2023

FILTRE

$\delta(t)$ → L.T.I. → $h(t)$

$u_i(t) \longrightarrow u_o(t) = u_i(t) * h(t)$

$h(t) \xrightarrow{\mathcal{L}} H(s)$

$u_i(t) \xrightarrow{\mathcal{L}} U_i(s)$

$u_o(t) \xrightarrow{\mathcal{L}} U_o(s)$

$U_o(s) = H(s)\, U_i(s)$

$u_o(t) = \mathcal{L}^{-1}\{U_o(s)\} = \mathcal{L}^{-1}(U_i(s)\, H(s))$

$u_i(t) = e^{j\omega t} \rightarrow |H(j\omega)|\, e^{j(\omega t + \angle H(j\omega))}$

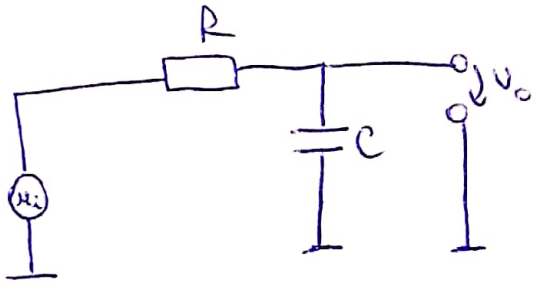
FILTRE ORD 1

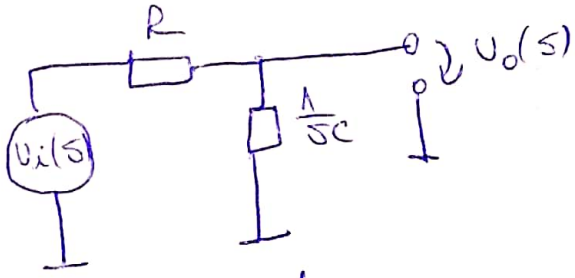
FTJ

FTS

(frecvență, pulsație)

$\omega = 2\pi f$





$U_o(s) = H(s) U_i(s)$

$H(s) = \frac{U_o(s)}{U_i(s)}$

$u_i(t) \longrightarrow U_i(s)$

$R \rightarrow R$

$C \rightarrow \frac{1}{sC}$

$L \rightarrow sL$

$U_o(s) = U_i(s) \cdot \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = U_i(s) \frac{1}{1 + RCs}$

$H(s) = \frac{U_o(s)}{U_i(s)} = \frac{1}{1 + RCs}$

$R = 1K$

$C = 1\mu F$

$u_i(t) = \sin(\underbrace{10^5}_{\omega} t)$

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$$u_o(t) = |H(j\cdot 10^5)| \sin(10^5 t + \angle H(j\cdot 10^5))$$

$$A = \left| \frac{1}{1 + 10^3 \cdot 10^{-6} \cdot 10^5 j} \right| = \left| \frac{1}{1 + 100j} \right| \approx \frac{1}{100} = -40\,dB$$

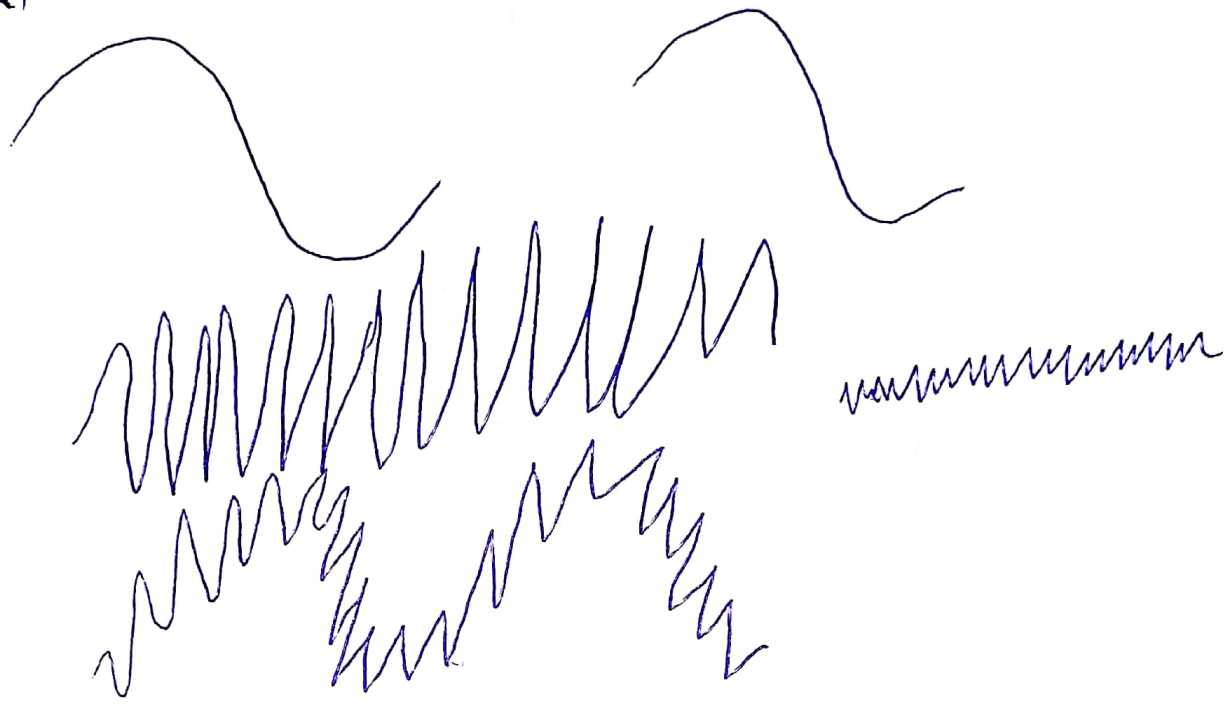
$R = 1K$

$C = 1\mu F$

$u_i = \sin(t)$

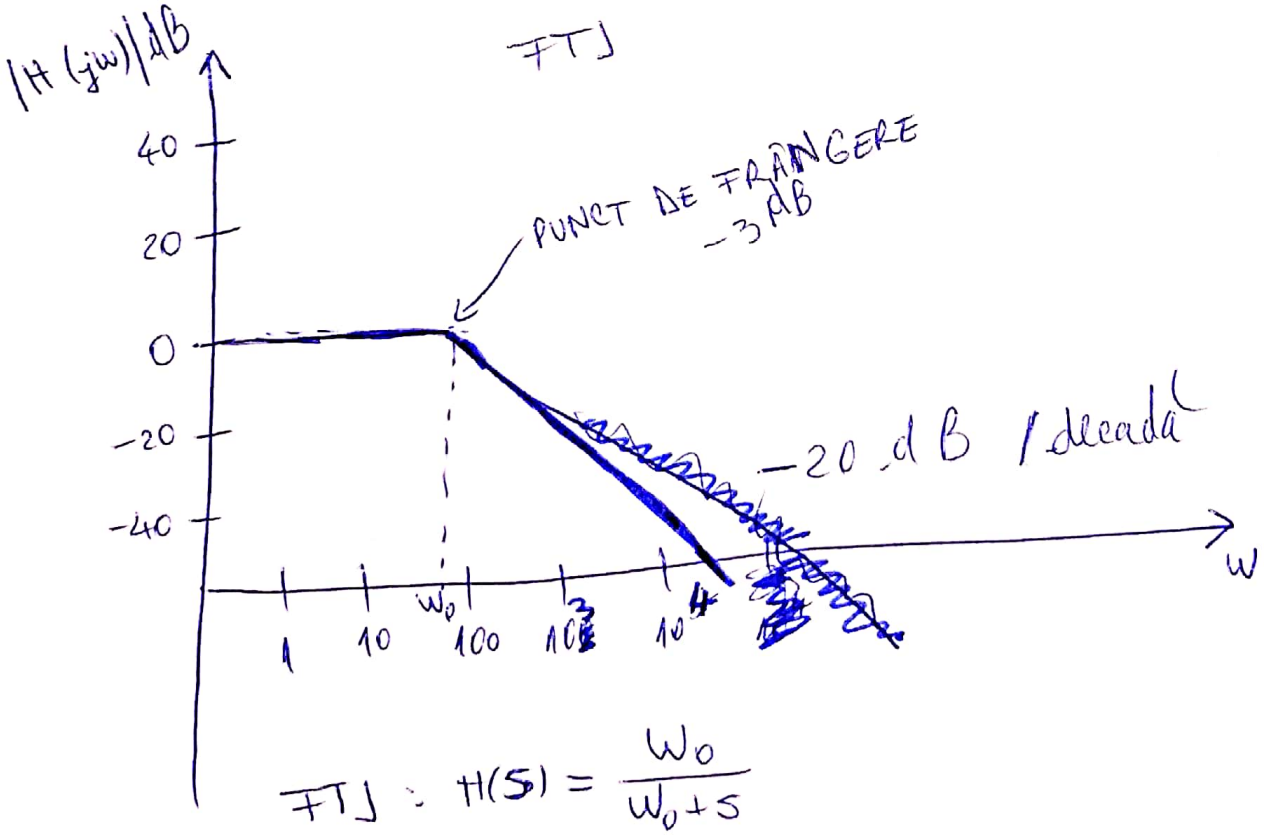
$$A = \left| \frac{1}{1 + 10^3 \cdot 10^{-6} j} \right| \approx 1$$

u_{i1}
$\omega = 1$



u_{i2}
$\omega = 10^5$

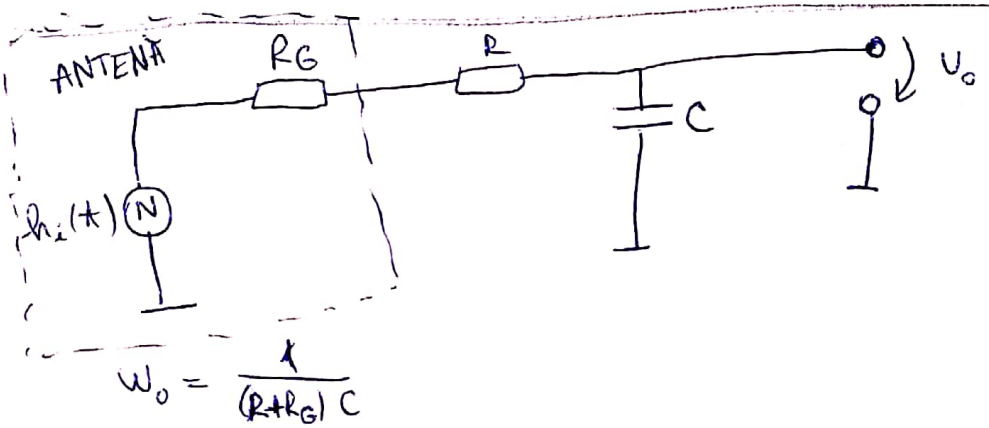
$u_{i1} + u_{i2}$



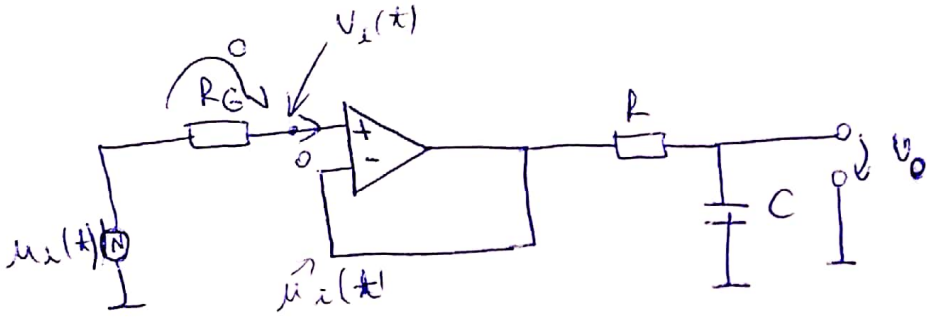
FTJ : $H(s) = \frac{w_0}{w_0 + s}$

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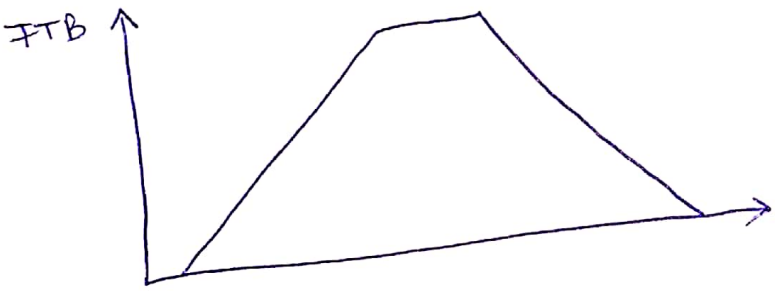
$$H(S) = \frac{1}{1+RCS} = \frac{\frac{1}{RC}}{\frac{1}{RC}+S} \qquad W_0 = \frac{1}{RC}$$

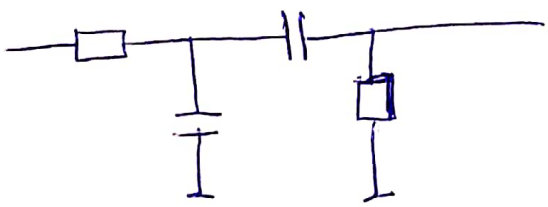


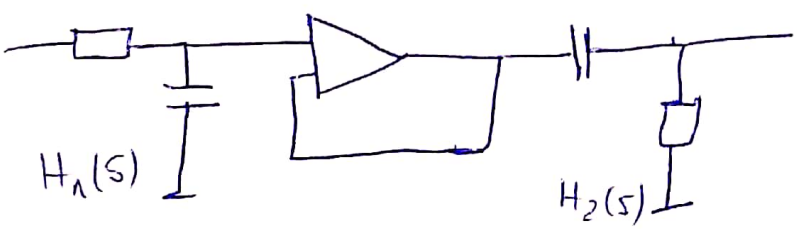
$$W_0 = \frac{1}{(R+R_G)\,C}$$



$$W_0 = \frac{1}{RC}$$

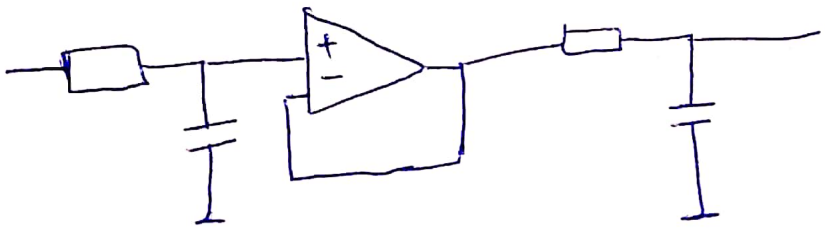






$$V_0(S) = H_1(S) \cdot H_2(S) \cdot U_i(S)$$

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TF j ord 2 $\left(\frac{w_0}{w_0+s}\right)^2$ 40 dB/dec

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